

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-Q

(Mark One)

☒

QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the quarterly period ended September 30, 2024

OR

☐

TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the transition period from to
Commission file number: 001-36481

ASPEN AEROGELS, INC.

(Exact name of registrant as specified in its charter)

Delaware
(State or other jurisdiction of
incorporation or organization)

04-3559972
(I.R.S. Employer
Identification No.)

30 Forbes Road, Building B
Northborough, Massachusetts
(Address of principal executive offices)

01532
(Zip Code)

Registrant’s telephone number, including area code: (508) 691-1111

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol	Name of exchange on which registered
Common Stock, par value \$0.00001 per share	ASPN	The New York Stock Exchange

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days.
Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of “large accelerated filer,” “accelerated filer,” “smaller reporting company,” and “emerging growth company” in Rule 12b-2 of the Exchange Act.

Large accelerated filer☐

Non-accelerated filer☐

Accelerated filer☒

Smaller reporting company☐

Emerging growth company☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of November 5, 2024, the registrant had 82,047,288 shares of common stock outstanding.

ASPEN AEROGELS, INC.

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Trademarks, Trade Names and Service Marks

We own or have rights to use “Aspen Aerogels,” “Cryogel,” “Pyrogel,” “Spaceloft,” “PyroThin,” the Aspen Aerogels logo and other trademarks, service marks and trade names of Aspen Aerogels, Inc. appearing in this Quarterly Report on Form 10-Q. Solely for convenience, the trademarks, service marks and trade names referred to in this report are presented without the ® and TM symbols, but such references are not intended to indicate, in any way, that the owner thereof will not assert, to the fullest extent under applicable law, such owner’s rights to these trademarks, service marks and trade names. This report contains additional trademarks, service marks and trade names of other companies, which, to our knowledge, are the property of their respective owners.

PART I — FINANCIAL INFORMATION

Item 1. Financial Statements.

ASPEN AEROGELS, INC. Consolidated Balance Sheets (Unaudited)

	September 30, 2024	December 31, 2023
	(In thousands, except share and per share data)	
Assets		
Current assets:		
Cash and cash equivalents	\$ 113,489	\$ 139,723
Restricted cash	394	248
Accounts receivable, net of allowances of \$898 and \$230	115,199	69,995
Inventories	47,430	39,189
Prepaid expenses and other current assets	29,344	17,176
Total current assets	305,856	266,331
Property, plant and equipment, net	451,569	417,227
Operating lease right-of-use assets	20,373	17,212
Other long-term assets	4,789	2,278
Total assets	<u>\$ 782,587</u>	<u>\$ 703,048</u>
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	\$ 42,280	\$ 51,094
Accrued expenses	26,531	22,811
Deferred revenue	2,405	2,316
Finance obligation for sale and leaseback transactions	3,653	—
Operating lease liabilities	3,181	1,874
Long term debt - current portion	26,250	—
Total current liabilities	104,300	78,095
Revolving line of credit	42,735	—
Long term debt	93,674	—
Convertible note - related party	—	114,992
Finance obligation for sale and leaseback transactions long-term	10,486	—
Operating lease liabilities long-term	23,742	21,906
Total liabilities	274,937	214,993
Commitments and contingencies (Note 9)		
Stockholders' equity:		
Preferred stock, \$0.00001 par value; 5,000,000 shares authorized, no shares issued and outstanding at September 30, 2024 and December 31, 2023	—	—
Common stock, \$0.00001 par value; 250,000,000 shares authorized, 77,155,896 and 76,503,151 shares issued and outstanding at September 30, 2024 and December 31, 2023, respectively	—	—
Additional paid-in capital	1,179,239	1,161,657
Accumulated deficit	(671,589)	(673,602)
Total stockholders' equity	507,650	488,055
Total liabilities and stockholders' equity	<u>\$ 782,587</u>	<u>\$ 703,048</u>

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Operations
(Unaudited)

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2024	2023	2024	2023
	(In thousands, except share and per share data)			
Revenue	\$ 117,340	\$ 60,755	\$ 329,611	\$ 154,499
Cost of revenue	68,297	46,945	193,847	127,196
Gross profit	49,043	13,810	135,764	27,303
Operating expenses:				
Research and development	4,591	4,218	13,645	12,281
Sales and marketing	9,306	8,386	27,130	24,226
General and administrative	17,746	15,840	52,465	41,382
Impairment of equipment under development	—	—	2,702	—
Total operating expenses	31,643	28,444	95,942	77,889
Income (loss) from operations	17,400	(14,634)	39,822	(50,586)
Other income (expense)				
Interest expense, convertible note - related party	(1,469)	(1,938)	(7,550)	(2,424)
Interest income (expense)	(1,147)	1,313	(883)	5,532
Income from Employee Retention Credits	—	2,186	—	2,186
Loss on extinguishment of debt	(27,487)	—	(27,487)	—
Total other income (expense)	(30,103)	1,561	(35,920)	5,294
Income (loss) before income tax expense	(12,703)	(13,073)	3,902	(45,292)
Income tax expense	(267)	—	(1,889)	—
Net income (loss)	\$ (12,970)	\$ (13,073)	\$ 2,013	\$ (45,292)
Net income (loss) per share:				
Basic	\$ (0.17)	\$ (0.19)	\$ 0.03	\$ (0.65)
Diluted	\$ (0.17)	\$ (0.19)	\$ 0.03	\$ (0.65)
Weighted-average common shares outstanding:				
Basic	76,261,294	69,317,805	76,402,123	69,243,843
Diluted	76,261,294	69,317,805	79,149,193	69,243,843

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Stockholders' Equity
(Unaudited)
(In thousands, except share data)

	Preferred Stock		Common Stock		Additional Paid-in Capital	Accumulated Deficit	Total Stockholders' Equity
	Shares	Value	Shares	Value			
Balance at December 31, 2023	—	\$ —	76,503,151	\$ —	\$ 1,161,657	\$ (673,602)	\$ 488,055
Net loss	—	—	—	—	—	(1,835)	(1,835)
Stock-based compensation expense	—	—	—	—	2,532	—	2,532
Issuance costs from private placement of common stock	—	—	—	—	(28)	—	(28)
Vesting of restricted stock units	—	—	118,289	—	(1,081)	—	(1,081)
Cancellation of restricted stock	—	—	(679,796)	—	2,174	—	2,174
Proceeds from employee stock option exercises	—	—	136,286	—	1,386	—	1,386
Balance at March 31, 2024	—	\$ —	76,077,930	\$ —	\$ 1,166,640	\$ (675,437)	\$ 491,203
Net income	—	—	—	—	—	16,818	16,818
Stock-based compensation expense	—	—	—	—	2,971	—	2,971
Issuance of restricted stock awards	—	—	11,388	—	—	—	—
Employee restricted stock awards withheld for tax	—	—	(75,885)	—	(1,810)	—	(1,810)
Vesting of restricted stock units	—	—	3,790	—	(53)	—	(53)
Proceeds from employee stock option exercises	—	—	1,063,816	—	8,698	—	8,698
Balance at June 30, 2024	—	\$ —	77,081,039	\$ —	\$ 1,176,446	\$ (658,619)	\$ 517,827
Net income	—	—	—	—	—	(12,970)	(12,970)
Stock-based compensation expense	—	—	—	—	2,630	—	2,630
Vesting of restricted stock units	—	—	11,784	—	(124)	—	(124)
Proceeds from employee stock option exercises	—	—	63,073	—	287	—	287
Balance at September 30, 2024	—	\$ —	77,155,896	\$ —	\$ 1,179,239	\$ (671,589)	\$ 507,650

	Preferred Stock		Common Stock		Additional Paid-in Capital	Accumulated Deficit	Total Stockholders' Equity
	Shares	Value	Shares	Value			
Balance at December 31, 2022	—	\$ —	69,994,963	\$ —	\$ 1,075,226	\$ (627,791)	\$ 447,435
Net loss	—	—	—	—	—	(16,796)	(16,796)
Stock-based compensation expense	—	—	—	—	2,267	—	2,267
Vesting of restricted stock units	—	—	71,643	—	(385)	—	(385)
Proceeds from employee stock option exercises	—	—	2,554	—	21	—	21
Balance at March 31, 2023	—	\$ —	70,069,160	\$ —	\$ 1,077,129	\$ (644,587)	\$ 432,542
Net loss	—	—	—	—	—	(15,423)	(15,423)
Stock-based compensation expense	—	—	—	—	2,710	—	2,710
Issuance of restricted stock	—	—	44,928	—	—	—	—
Vesting of restricted stock units	—	—	2,464	—	(8)	—	(8)
Proceeds from employee stock option exercises	—	—	41,591	—	150	—	150
Balance at June 30, 2023	—	\$ —	70,158,143	\$ —	\$ 1,079,981	\$ (660,010)	\$ 419,971
Net loss	—	—	—	—	—	(13,073)	(13,073)
Stock-based compensation expense	—	—	—	—	2,789	—	2,789
Vesting of restricted stock units	—	—	2,662	—	(9)	—	(9)
Proceeds from employee stock option exercises	—	—	65,828	—	274	—	274
Issuance costs from underwritten public offering	—	—	—	—	(139)	—	(139)
Balance at September 30, 2023	—	\$ —	70,226,633	\$ —	\$ 1,082,896	\$ (673,083)	\$ 409,813

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Cash Flows
(Unaudited)

	Nine Months Ended September 30,	
	2024	2023
	(In thousands)	
Cash flows from operating activities:		
Net income (loss)	\$ 2,013	\$ (45,292)
Adjustments to reconcile net income (loss) to net cash used in operating activities:		
Depreciation	17,093	10,757
Accretion of interest on convertible note - related party	7,083	1,721
Amortization of convertible note issuance costs	320	28
Amortization of debt discount due to modification of convertible note – related party	443	675
Loss on extinguishment of debt	27,487	—
Deferred financing costs written off	1,829	—
Provision for bad debt	139	(89)
Stock-based compensation expense	10,307	7,766
Impairment of property, plant and equipment	6,810	—
Reduction in the carrying amount of operating lease right-of-use assets	1,990	2,186
Changes in operating assets and liabilities:		
Accounts receivable	(45,343)	3,026
Inventories	(8,241)	(11,883)
Prepaid expenses and other assets	(16,786)	(6,771)
Accounts payable	4,452	(420)
Accrued expenses	1,890	691
Deferred revenue	89	(383)
Operating lease liabilities	(1,710)	(1,845)
Net cash provided (use) in operating activities	9,865	(39,833)
Cash flows from investing activities:		
Capital expenditures	(71,511)	(147,669)
Net cash used in investing activities	(71,511)	(147,669)
Cash flows from financing activities:		
Proceeds from employee stock option exercises	10,371	445
Proceeds from sale and leaseback transactions	14,982	—
Repayment of finance obligation for sale and leaseback transactions	(843)	—
Payments made for employee restricted stock tax withholdings	(1,258)	(402)
Repayment of convertible note	(150,029)	—
Proceeds from term loan	125,000	—
Issuance costs from term loan	(5,337)	—
Proceeds from revolver	43,000	—
Issuance costs from revolver	(300)	—
Fees and issuance costs from private placement of common stock	(28)	(139)
Net cash provided by (used in) financing activities	35,558	(96)
Net decrease in cash, cash equivalents and restricted cash	(26,088)	(187,598)
Cash, cash equivalents and restricted cash at beginning of period	139,971	282,561
Cash, cash equivalents and restricted cash at end of period	\$ 113,883	\$ 94,963
Supplemental disclosures of cash flow information:		
Interest paid	\$ 2,328	\$ 1
Income taxes paid	\$ —	\$ —
Supplemental disclosures of non-cash activities:		
Right-of-use assets obtained in exchange for new operating lease liabilities	\$ 4,853	\$ 9,994
Capitalized interest	\$ —	\$ 6,084
Changes in accrued capital expenditures	\$ (13,266)	\$ (17,193)

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS
(UNAUDITED)

(1) Description of Business and Basis of Presentation

Nature of Business

Aspen Aerogels, Inc. (the Company) is an aerogel technology company that has introduced a line of aerogel thermal barriers for use in battery packs in the electric vehicle market. In addition, the Company designs, develops and manufactures innovative, high-performance aerogel insulation used in the energy industrial and sustainable insulation materials markets. The Company is also developing applications for its aerogel technology in the battery materials and a number of other high-potential markets.

The Company maintains its corporate offices in Northborough, Massachusetts. The Company has three wholly owned subsidiaries: Aspen Aerogels Rhode Island, LLC (Aspen RI), Aspen Aerogels Germany, GmbH and Aspen Aerogels Georgia, LLC (Aspen Georgia). Additionally, we engaged Prodensa Servicios de Consultora to establish OPE Manufacturer Mexico S de RL de CV, a maquiladora located in Mexico (OPE), which manufactures thermal barrier PyroThin products and operates an automated fabrication facility for PyroThin. OPE is currently owned by Prodensa, which charges a management fee. There is an option for OPE to be purchased by the Company after a period of 18 months. During the period between inception and the exercise of the purchase option, OPE operations are consolidated within the Company financial statements.

Liquidity

During the nine months ended September 30, 2024, the Company earned net income of \$2.0 million, provided \$9.9 million of cash in operations and used \$71.5 million of cash for capital expenditures. The Company had unrestricted cash and cash equivalents of \$113.5 million as of September 30, 2024.

In January 2024 and September 2024, the Company entered into sale and leaseback arrangements, pursuant to which the Company sold certain equipment to an equipment leasing company for one-time cash payments of \$5.0 and \$10.0 million, respectively, and leased back such equipment from the leasing company. The associated monthly lease rents will be paid over the lease term of three years.

On August 19, 2024, the Company and Aspen RI (each, a Borrower and collectively, the Borrowers) entered into a Credit, Security and Guaranty Agreement (the Credit Agreement and the facilities provided thereunder, collectively, the MidCap Loan Facility) with MidCap Funding IV Trust, as agent (the Agent), MidCap Financial Trust, as term loan servicer, the financial institutions or other entities from time to time party thereto as lenders (the Lenders), and the other parties party thereto as additional guarantors and/or borrowers from time to time, for a term loan facility in an aggregate principal amount of \$125.0 million and an asset-based revolving credit facility in an aggregate principal amount not to exceed the lesser of \$100.0 million or the borrowing base. At closing the Company drew \$125.0 million from the term loan facility and \$43.0 million from the revolving loan facility. The proceeds of the borrowings, net of fees and costs, were used for repurchasing of the 2022 Convertible Notes (as defined below) for \$150.0 million and for general corporate purposes. The amount available to the Company at September 30, 2024 under the revolving loan facility was \$25.7 million.

The Company is increasing investment in the research and development of next-generation aerogel products and manufacturing process technologies. In addition, the Company has developed a number of promising aerogel products and technologies for the electric vehicle market. The Company believes that the commercial potential for the Company's products and technology in the electric vehicle market is significant. Accordingly, the Company is hiring additional personnel, incurring additional operating expenses, incurring significant capital expenditures to expand aerogel manufacturing capacity and automated thermal barrier fabrication operations, and enhancing research and development resources, among other items.

The Company expects its existing cash balance and the amount anticipated to be available under the existing revolving loan facility will be sufficient to support current operating requirements, current research and development activities and the initial capital expenditures required to support the evolving commercial opportunity in the electric vehicle market and other strategic business initiatives. However, the Company plans to supplement its cash balance and available credit with equity financings, debt financings, equipment leasing, sale and leaseback transactions, customer prepayments, or government grant and loan programs to provide the

additional capital necessary to purchase the capital equipment, construct the new facilities, establish the operations and complete the aerogel capacity expansions required to support these evolving commercial opportunities and strategic business initiatives.

Unaudited Interim Financial Information

The accompanying unaudited interim consolidated financial statements include the accounts of the Company and have been prepared in accordance with accounting principles generally accepted in the United States of America (U.S. GAAP) and with the instructions to Form 10-Q and Article 10 of Regulation S-X. Certain information and disclosures normally included in the consolidated financial statements prepared in accordance with U.S. GAAP have been condensed or omitted pursuant to such rules and regulations. As such, the information included in this Quarterly Report on Form 10-Q should be read in conjunction with the audited consolidated financial statements and accompanying notes in our Annual Report on Form 10-K for the year ended December 31, 2023 (the Annual Report), filed with the U.S. Securities and Exchange Commission on March 7, 2024.

In the opinion of the Company's management, the unaudited interim consolidated financial statements have been prepared on the same basis as the audited consolidated financial statements and include all adjustments that are of a normal recurring nature and necessary for the fair statement of the Company's financial position as of September 30, 2024 and the results of its operations and stockholders' equity for the three and nine months ended September 30, 2024 and 2023 and the cash flows for the nine-month periods then ended. The Company has evaluated subsequent events through the date of this filing.

The Company's results of operations for the three and nine months ended September 30, 2024 are not necessarily indicative of the results to be expected for the year ending December 31, 2024 or any other period.

(2) Significant Accounting Policies

Please refer to "Note 2. Summary of Basis of Presentation and Significant Accounting Policies," to the Company's consolidated financial statements from the Annual Report for the discussion of the Company's significant accounting policies.

Use of Estimates

The preparation of the consolidated financial statements requires the Company to make a number of estimates and assumptions that affect the reported amounts of assets and liabilities at the date of the consolidated financial statements and the reported amounts of revenues and expenses during the reporting period. Significant items subject to such estimates and assumptions include allowances for doubtful accounts, sales returns and allowances, product warranty costs, inventory valuation, the carrying amount of property and equipment, right-of-use assets, lease liabilities, stock-based compensation, and deferred income taxes. The Company evaluates its estimates and assumptions on an on-going basis using historical experience and other factors, including current economic conditions, which are believed to be reasonable under the circumstances. Management adjusts such estimates and assumptions when facts and circumstances warrant. Illiquid credit markets, volatile equity markets and declines in business investment can increase the uncertainty inherent in such estimates and assumptions. As future events and their effects cannot be determined with precision, actual results could differ significantly from these estimates. Changes in these estimates resulting from continuing changes in the economic environment will be reflected in the financial statements in future periods.

Restricted Cash

As of September 30, 2024, the Company had \$0.4 million of restricted cash to support its outstanding letters of credit.

Concentration of Credit Risk

Financial instruments, which potentially expose the Company to concentrations of credit risk, consist principally of accounts receivable. The Company's customers are primarily insulation distributors, insulation contractors, insulation fabricators and select energy and automotive end-users located throughout the world. The Company performs ongoing credit evaluations of its customers' financial condition and generally requires no collateral to secure accounts receivable. The Company maintains an allowance for doubtful accounts based on its assessment of the collectability of accounts receivable. The Company reviews the allowance for doubtful accounts quarterly. During the nine months ended September 30, 2024, the Company recorded an increase for estimated customer uncollectible accounts receivable of \$0.1 million. During the nine months ended September 30, 2023, the Company recorded a reduction for estimated customer uncollectible accounts receivable of less than \$0.1 million.

For the nine months ended September 30, 2024 and 2023, two customers represented 74% and 50% of total revenue, respectively.

At September 30, 2024, the Company had one customer which accounted for 79% of accounts receivable. At December 31, 2023, the Company had two customers which accounted for 60% and 6% of accounts receivable, respectively.

Revenue Recognition

The Company recognizes revenue in accordance with Accounting Standards Codification 606, Revenue from Contracts with Customers (ASC 606). See note 3 for further details.

Sale and Leaseback Accounting

The Company has entered into sale and leaseback transactions for certain equipment within its plants. Due to the Company not meeting criteria to account for the transfer of the assets as a sale, sale accounting is precluded. Accordingly, the Company uses the financing method to account for these transactions.

Under the financing method of accounting for a sale and leaseback, the Company does not derecognize the assets and does not recognize as revenue any of the sale proceeds received from the lessor that contractually constitutes payment to acquire the assets subject to these arrangements. Instead, the sale proceeds received are accounted for as finance obligations and leaseback payments made by the Company are allocated between interest expense and a reduction to the finance obligation. Interest on the finance obligation is calculated using the Company's incremental borrowing rate at the inception of the arrangement on the outstanding finance obligation.

Recently Issued Accounting Standards

From time to time, new accounting pronouncements are issued by the Financial Accounting Standards Board or other standard setting bodies. Recently issued standards typically do not require adoption until a future effective date. Prior to their effective date, the Company evaluates the pronouncements to determine the potential effects of adoption to its consolidated financial statements.

Standards Implemented Since December 31, 2023

The Company has not implemented any accounting standards that had a material impact on its consolidated financial statements during the nine months ended September 30, 2024.

Standards to be Implemented

In November 2023, the Financial Accounting Standards Board (FASB) issued Accounting Standard Update (ASU) 2023-07 Segment Reporting (Topic 280) Improvements to Reportable Segment Disclosures to enhance disclosures about significant segment expenses. This ASU is effective for the Company's fiscal year 2024 and interim periods in fiscal year 2025. Early adoption is permitted. The Company is currently evaluating segment expense disclosures related to its annual report for fiscal year 2024.

In December 2023, the FASB issued ASU 2023-09 Income Taxes (Topic 740) Improvements to Income Tax Disclosures that requires disclosure of disaggregated income taxes paid, prescribes standard categories for the components of the effective tax rate reconciliation, and modifies other income tax-related disclosures. This ASU is effective for the Company's fiscal year 2025. Early adoption is permitted. The Company is currently evaluating income tax disclosures related to its annual report for fiscal year 2025.

Although there are several other new accounting pronouncements issued by the FASB, the Company does not believe any of these accounting pronouncements had or will have a material impact on its Consolidated Financial Statements.

The Company believes that the impact of recently issued accounting standards that are not yet effective will not have a material impact on its consolidated financial statements.

(3) Revenue from Contracts with Customers

Revenue Recognition

Revenue is recognized when a customer obtains control of promised goods or services in an amount that reflects the consideration which the entity expects to receive in exchange for those goods or services. To determine revenue recognition for arrangements within the scope of ASC 606, the Company performs the following five steps: (i) identification of the contract with a customer; (ii) identification of the performance obligations in the contract; (iii) determination of the transaction price; (iv) allocation of the transaction price to the separate performance obligations in the contract; and (v) recognition of the revenue associated with performance obligations as they are satisfied. The Company applies the five-step model to contracts when it is probable that the Company will collect the consideration it is entitled to in exchange for the goods or services it transfers to the customer. At contract inception, once the contract is determined to be within the scope of ASC 606, the Company assesses the goods or services promised within each contract and determines those that are performance obligations and assesses whether each promised good or service is distinct. If the contract contains a single performance obligation, the entire transaction price is allocated to the single performance obligation. Contracts that contain multiple performance obligations require an allocation of the transaction price based on the estimated relative standalone-selling prices of the promised products or services underlying each performance obligation. The Company determines standalone-selling prices based on the price at which the performance obligation is sold separately. If the standalone-selling price is not observable through past transactions, the Company estimates the standalone-selling price considering available information such as market conditions and internally approved pricing guidelines related to the performance obligations. The Company then recognizes as revenue the amount of the transaction price that is allocated to the respective performance obligation when (or as) the performance obligation is satisfied.

When determining the transaction price of a contract, an adjustment is made if payment from a customer occurs either significantly before or significantly after performance, resulting in a significant financing component. Applying the practical expedient in paragraph ASC 606-10-32-18, the Company does not assess whether a significant financing component exists if the period between when the Company performs its obligations under the contract and when the customer pays is one year or less. The Company did not have any contracts outstanding at December 31, 2023 and did not enter into any contracts during the nine months ended September 30, 2024 that contained a significant financing component.

The Company records deferred revenue for product sales when (i) the Company has delivered products, but other revenue recognition criteria have not been satisfied, or (ii) payments have been received in advance of the completion of required performance obligations.

Thermal Barriers

The Company supplies fabricated, multi-part thermal barriers for use in battery packs in the electric vehicle market. These thermal barriers are customized to meet customer specifications. Although thermal barrier products are customized with no alternative use to the Company, the Company does not always have an enforceable right to payment. Under the provisions of ASC 606, the Company recognizes revenue at a point in time when transfer of the control of the products is passed to the customer according to the terms of the contract, including under bill and hold arrangements. The timing of revenue recognition is assessed on a contract-by-contract basis.

Energy Industrial

The Company generally enters into contracts containing one type of performance obligation. For a majority of the contracts, the Company recognizes revenue at a point in time when transfer of control of the products is passed to the customer, which is generally upon delivery according to contractual shipping terms within customer purchase orders. For a limited number of customer arrangements for customized products with no alternative use to the Company and an enforceable right to payment for progress completed to date, the Company recognizes revenue over time using units of production to measure progress toward satisfying the performance obligations. Units of production represent work performed as we do not generate significant work in process and thereby best depicts the transfer of control to the customer. Customer invoicing terms for contracts for which revenue is recognized under the over time methodology are typically based on certain milestones within the production and delivery schedule. The timing of revenue recognition is assessed on a contract-by-contract basis.

The Company also enters into rebate agreements with certain customers. These agreements may be considered an additional performance obligation of the Company or variable consideration within a contract. Rebates are recorded as a reduction of revenue in

the period the related revenue is recognized. A corresponding liability is recorded as a component of deferred revenue on the consolidated balance sheets. These arrangements are primarily based on the customer attaining contractually specified sales volumes.

The Company estimates the amount of its sales that may be returned by its customers and records this estimate as a reduction of revenue in the period the related revenue is recognized. The Company currently estimates return liabilities using historical rates of return, current quarter credit sales, and specific items of exposure on a contract-by-contract basis. Sales return reserves were approximately \$0.7 million and \$0.2 million as of September 30, 2024 and December 31, 2023, respectively.

Shipping and Handling Costs

Shipping and handling costs associated with outbound freight after control over a product has transferred to a customer are accounted for as fulfillment costs and are included in the cost of product revenue. The associated amount of revenue recognized includes the consideration to which the Company expects to be entitled to receive in exchange for incurring these shipping and handling costs.

Disaggregation of Revenue

In the following tables, revenue is disaggregated by primary geographical region and source of revenue:

	Three Months Ended September 30,					
	2024			2023		
	U.S.	International	Total	U.S.	International	Total
	(In thousands)					
Geographical region						
Asia	\$ —	\$ 4,439	\$ 4,439	\$ —	\$ 5,801	\$ 5,801
Canada	—	6,125	6,125	—	633	633
Europe	—	8,455	8,455	—	11,361	11,361
Latin America	—	25,012	25,012	—	1,176	1,176
U.S.	73,309	—	73,309	41,784	—	41,784
Total revenue	<u>\$ 73,309</u>	<u>\$ 44,031</u>	<u>\$ 117,340</u>	<u>\$ 41,784</u>	<u>\$ 18,971</u>	<u>\$ 60,755</u>
Source of revenue						
Energy industrial	\$ 13,679	\$ 13,097	\$ 26,776	\$ 12,249	\$ 15,663	\$ 27,912
Thermal barrier	59,630	30,934	90,564	29,535	3,308	32,843
Total revenue	<u>\$ 73,309</u>	<u>\$ 44,031</u>	<u>\$ 117,340</u>	<u>\$ 41,784</u>	<u>\$ 18,971</u>	<u>\$ 60,755</u>

	Nine Months Ended September 30,					
	2024			2023		
	U.S.	International	Total	U.S.	International	Total
	(In thousands)					
Geographical region						
Asia	\$ —	\$ 18,071	\$ 18,071	\$ —	\$ 27,522	\$ 27,522
Canada	—	11,599	11,599	—	1,519	1,519
Europe	—	26,926	26,926	—	26,735	26,735
Latin America	—	74,614	74,614	—	5,065	5,065
U.S.	198,401	—	198,401	93,658	—	93,658
Total revenue	<u>\$ 198,401</u>	<u>\$ 131,210</u>	<u>\$ 329,611</u>	<u>\$ 93,658</u>	<u>\$ 60,841</u>	<u>\$ 154,499</u>
Source of revenue						
Energy industrial	\$ 41,853	\$ 50,928	\$ 92,781	\$ 43,994	\$ 53,317	\$ 97,311
Thermal barrier	156,548	80,282	236,830	49,664	7,524	57,188
Total revenue	<u>\$ 198,401</u>	<u>\$ 131,210</u>	<u>\$ 329,611</u>	<u>\$ 93,658</u>	<u>\$ 60,841</u>	<u>\$ 154,499</u>

Contract Balances

The following table presents changes in the Company's contract liabilities during the nine months ended September 30, 2024:

	Balance at December 31, 2023	Additions	Deductions	Balance at September 30, 2024
(In thousands)				
Contract liabilities				
Deferred revenue				
Energy industrial	\$ 2,316	\$ 5,686	\$ (5,597)	\$ 2,405
Total contract liabilities	<u>\$ 2,316</u>	<u>\$ 5,686</u>	<u>\$ (5,597)</u>	<u>\$ 2,405</u>

During the nine months ended September 30, 2024, the Company recognized \$2.3 million of revenue that was included in deferred revenue as of December 31, 2023.

A contract asset is recorded when the Company satisfies a performance obligation by transferring a promised good or service and has earned the right to consideration from its customer. These assets may represent a conditional right to consideration and are included within accounts receivable and other current assets on the consolidated balance sheets.

A contract liability is recorded when consideration is received, or such consideration is unconditionally due, from a customer prior to transferring goods or services under the terms of the contract. Contract liabilities are recognized as revenue after control of the products or services is transferred to the customer and all revenue recognition criteria have been met.

(4) Inventories

Inventories consist of the following:

	September 30, 2024	December 31, 2023
(In thousands)		
Raw materials	\$ 16,054	\$ 24,735
Work in process	12,226	7,936
Finished goods	19,150	6,518
Total	<u>\$ 47,430</u>	<u>\$ 39,189</u>

(5) Property, Plant and Equipment, Net

Property, plant and equipment consist of the following:

	September 30, 2024	December 31, 2023	Useful life
(In thousands)			
Construction in progress	\$ 344,063	\$ 314,695	—
Buildings	26,914	25,473	30 years
Machinery and equipment	197,657	185,339	3-10 years
Computer equipment and software	9,703	9,495	3 years
Leasehold improvements	24,570	23,514	Shorter of useful life or lease term
Total	<u>602,907</u>	<u>558,516</u>	
Accumulated depreciation	<u>(151,338)</u>	<u>(141,289)</u>	
Property, plant and equipment, net	<u>\$ 451,569</u>	<u>\$ 417,227</u>	

Depreciation expense was \$17.1 million and \$10.8 million for the nine months ended September 30, 2024 and 2023, respectively.

The Company recorded impairment charges of approximately \$6.8 million during the nine months ended September 30, 2024, respectively, for equipment that will no longer be needed in manufacturing following customer directed engineering changes to a part it manufactures and for other property, plant and equipment that have become obsolete following development of new and more efficient equipment. Refer to Note 9 – Commitments and Contingencies for a discussion of the claim that the Company has submitted with a customer for reimbursement of losses incurred in connection with the customer directed engineering changes. The impairment charges of \$6.8 million during the nine months ended September 30, 2024 consist of \$4.1 million impairment included in cost of revenue and \$2.7 million included in impairment of equipment under development on the Company's consolidated statement of operations. There were no impairments of property, plant and equipment during the nine months ended September 30, 2023.

The construction in progress balance at September 30, 2024 and December 31, 2023 included engineering designs and construction costs, and capitalized interest totaling \$325.1 million and \$288.5 million, respectively, for a planned aerogel manufacturing facility in Bulloch County, Georgia. The Company incurred \$8.8 million in capitalized interest for the construction in progress in Bulloch County, Georgia. The Company capitalized interest of \$0.0 million and \$8.8 million for the nine months ended September 30, 2024 and 2023, respectively.

(6) Accrued Expenses

Accrued expenses consist of the following:

	September 30, 2024	December 31, 2023
	(In thousands)	
Employee compensation	\$ 15,003	\$ 16,876
Other accrued expenses	11,528	5,935
Total	<u>\$ 26,531</u>	<u>\$ 22,811</u>

(7) Related Party Transactions

Convertible Note

During the year ended December 31, 2022, the Company issued a \$100.0 million aggregate principal amount convertible note (the 2022 Convertible Note) to Wood River Capital, LLC (Wood River), an entity affiliated with Koch Disruptive Technologies, LLC (Koch), for the planned manufacturing facility in Bulloch County, Georgia, pursuant to that certain Note Purchase Agreement, dated as of February 15, 2022, as amended by that certain Amendment No. 1 to Note Purchase Agreement, dated November 28, 2022 (together, the Note Purchase Agreement).

The Company repurchased the 2022 Convertible Note on August 19, 2024. See Note (8) Convertible Note – Related Party for additional detail.

During the nine months ended September 30, 2024, the Company incurred \$7.1 million of interest from the 2022 Convertible Note.

Other

The Company had \$2.8 million in accounts payable as of December 31, 2023, due to an entity affiliated with Koch for project management service. On March 27, 2024, we entered into a Settlement and Release Agreement with the affiliate of Koch to settle the accounts payable for \$1.2 million, which was paid during the three months ended, June 30, 2024.

(8) Convertible Note – Related Party

2022 Convertible Note

On February 15, 2022, the Company entered into the Note Purchase Agreement with Wood River, an entity affiliated with Koch, relating to the issuance and sale to Wood River of the 2022 Convertible Note in the aggregate principal amount of \$100.0 million. The transactions contemplated by the Note Purchase Agreement closed on February 18, 2022. The maturity date of the 2022 Convertible Note is February 18, 2027, subject to earlier conversion, redemption, or repurchase.

The 2022 Convertible Note is a senior unsecured obligation of the Company and ranks equal in right of payment to all senior unsecured indebtedness of the Company and will rank senior in right of payment to any indebtedness that is contractually subordinated to the 2022 Convertible Note.

On August 19, 2024, the Company entered into a note purchase and sale agreement with Wood River (the Note Repurchase Agreement), pursuant to which the Company repurchased from Wood River \$123.9 million in aggregate capitalized principal amount (inclusive of PIK interest paid through June 30, 2024) of the 2022 Convertible Note, such aggregate amount being the entire outstanding amount of the 2022 Convertible Note, for a total purchase price of \$150.1 million in cash, which amount equals to the Redemption Price (as defined in the 2022 Convertible Note). Pursuant to the Note Repurchase Agreement, all rights and obligations, covenants and agreements under the 2022 Convertible Note and the Note Purchase Agreement were satisfied and discharged. The Redemption Price less capitalized principal amount and accrued interest to redemption date, of \$24.6 million along with unamortized deferred issuance costs was classified in the income statement as Loss on Extinguishment of Debt.

In accordance with ASU 2020-06, the 2022 Convertible Note is accounted for as a single unit of account and consists of the following:

	September 30, 2024	December 31, 2023
	(In thousands)	
Convertible note, principal	\$ —	\$ 100,000
Payment in-kind	—	18,318
Discount on convertible note, net of accumulated amortization	—	(3,209)
Debt issuance costs, net of accumulated amortization	—	(117)
Convertible note	<u>\$ —</u>	<u>\$ 114,992</u>

The 2022 Convertible Note does not have current observable inputs such as recent trading prices (Level 1) and was measured at fair value using a combination of option pricing and discounted cash flow models and incorporate management's assumptions for stock price, volatility and risk rate. In general, fair values determined by Level 1 inputs utilize observable inputs such as quoted prices in active markets for identical assets or liabilities. Fair values determined by Level 2 inputs utilize data points that are either directly or indirectly observable, such as quoted prices for similar instruments in active markets, interest rates and yield curves. Fair values determined by Level 3 inputs utilize unobservable data points in which there is little or no market data, which require the Company to develop its own assumptions for the asset or liability.

The Company estimated the fair value of the 2022 Convertible Note was approximately \$118.2 million as of December 31, 2023. However, as the Company did not elect to utilize the fair value option, it was carried at amortized cost of \$115.0 million.

(9) Debt

On August 19, 2024, the Borrowers entered into the Credit Agreement, by and among the Borrowers, the Agent, the Lenders and the other parties party thereto as additional guarantors and/or borrowers from time to time. Loans borrowed under the MidCap Loan Facility mature on August 19, 2029.

The MidCap Loan Facility is comprised of (i) a term loan facility in an aggregate principal amount of \$125 million (the Term Loan Facility) and (ii) an asset-based revolving credit facility in an aggregate principal amount not to exceed the lesser of (A) a \$100 million commitment amount and (B) the value of the borrowing base (defined as the sum of (x) 85% of certain eligible accounts of the Borrowers and (y) the lesser of 85% of the NOLV or 85% of the cost of certain eligible inventory of the Borrowers) (the "Revolving Facility"). Loans borrowed under the Term Loan Facility will bear interest rate equal to Term SOFR (as defined in the Credit Agreement) for a one-month interest period plus 4.50% per year, subject to a Term SOFR floor of 4.50% and a Term SOFR cap of 7.50%. Loans borrowed under the Revolving Facility will bear an interest rate equal to Term SOFR plus 4.60% per year, subject to a Term SOFR floor of 2.50%. The Term Loan Facility is subject to amortization of principal, payable quarterly on the last day of each quarter, commencing September 30, 2024, in an amount as set forth in the Credit Agreement with the remaining aggregate principal amount payable on the maturity date. The Borrowers are required to pay the Lenders an exit fee of \$2.5 million on the maturity date. The Revolving Facility has a required minimum balance set at 30% of the average borrowing base during the immediate preceding month. The Borrowers are required to pay the Lenders under the Revolving Facility an unused line fee of 0.30% of the average unused availability under the Revolving Facility, subject to the aforementioned minimum balance.

The MidCap Loan Facility is guaranteed by the Loan Parties and is secured by a lien on substantially all existing and after-acquired assets of the Loan Parties, including the equity interest in Aspen RI and Aspen Georgia, owned by the Company, in each

case, subject to customary exceptions. Aspen Georgia is not a guarantor and its assets are excluded from the collateral under the Credit Agreement, subject to its entrance into a loan facility within one year from the closing date of the MidCap Loan Facility. Once Aspen Georgia enters into the loan facility within such one-year period, the lien granted by the Company on the equity interest of Aspen Georgia will be released. If Aspen Georgia does not enter into the loan facility within such one-year period, Aspen Georgia will be obligated to become a Loan Party under the MidCap Loan Facility and pledge substantially all of its assets as security for the obligations thereunder.

The Credit Agreement includes representations and warranties, affirmative covenants (including reporting obligations), negative covenants and events of default that are usual and customary for facilities of this type, in each case, subject to certain permitted exceptions as set forth therein. The Credit Agreement includes financial covenants for the benefit of the Lenders, including (i) a covenant to maintain Liquidity (as defined therein) equal to or greater than \$75 million at all times and (ii) a covenant to maintain EBITDA (as defined therein) equal to or greater than the specified applicable amount set forth in the Credit Agreement, tested quarterly with the first test set at \$45 million commencing with the fiscal quarter ended September 30, 2024.

The Borrowers will have the right to prepay the loans outstanding under the MidCap Loan Facility (or, with respect to the Revolving Facility, terminate the commitments thereunder), subject to a premium equal to 3.0% of the amount prepaid or terminated, as applicable, during the first year after the closing date, which premium will be decreased to 2.0% during the second year after the closing date and to 1.0% thereafter. The Borrowers will be required to mandatorily prepay the loans outstanding under the Term Loan Facility with, among other things, certain casualty insurance proceeds or proceeds from non-ordinary course assets sales (which will also be subject to the aforementioned premium). The Borrowers will be required to mandatorily prepay the balance outstanding under the Revolving Facility (i) if the outstandings exceed the borrowing base in an aggregate amount equal to that excess or (ii) upon a cash dominion event of all the funds deposited in the lockbox account during the cash dominion period. A cash dominion event is triggered upon an event of default or if the Liquidity is less than \$100 million.

MidCap debt consists of the following:

	September 30, 2024 (In thousands)
Term loan	125,000
Exit fee	85
Term loan issuance costs	(5,161)
Total debt	119,924
Current portion	26,250
Long term portion	93,674

The revolving line of credit had an outstanding balance of \$42.7 million as of September 30, 2024, net of unamortized issuance costs.

(10) Commitments and Contingencies

Cloud Computing Agreement

The Company is party to multiple cloud computing agreements that are service contracts for enterprise resource planning software and payroll services. The amortization period of a cloud computing agreement was adjusted during the three months ended March 31, 2024 to align with implementation of a new agreement that will begin to amortize during 2025. The amortization associated with the payroll services agreement began during the three months ended September 30, 2024 and will amortize over a period of five years. The capitalized implementation costs are classified on the consolidated balance sheets as follows:

	September 30, 2024	December 31, 2023
	(In thousands)	
Cloud computing costs included in other current assets	\$ 780	\$ 420
Cloud computing costs included in other assets	2,691	1,590
Amortization of cloud computing costs	(578)	(662)
Total capitalized cloud computing costs	<u>\$ 2,893</u>	<u>\$ 1,348</u>

Thermal Barrier Contracts

The Company is party to production contracts with General Motors to supply fabricated, multi-part thermal barriers (Barriers) for use in the battery system of its next-generation electric vehicles (Contracts). Pursuant to the Contracts, the Company is obligated to supply Barriers at fixed annual prices and at volumes to be specified by General Motors up to a daily maximum quantity through the respective terms of the agreements, which expire at various times from 2026 through 2034. While General Motors has agreed to purchase its requirement for Barriers from the Company for locations to be designated from time to time by General Motors, it has no obligation to purchase any minimum quantity of Barriers under the Contracts. In addition, General Motors may terminate the Contracts at any time and for any or no reason. All other terms of the Contracts are generally consistent with General Motors' standard purchase terms, including quality and warranty provisions customary in automotive industry.

Charges for Engineering Change

In January 2024, the Company was notified by a customer of an engineering change to one of the parts the Company manufactures for that customer to enable incremental productivity and support a set of broader system level changes that could drive higher demand for its parts. The Company has submitted claims to the customer for reimbursement for estimated inventory and equipment losses incurred by the Company and its vendors due to potential obsolescence. In connection with the same, during the three months ended March 31, 2024, the Company recognized a charge of \$6.8 million, net of contractual recoverable of \$1.9 million, in cost of revenues for inventory obsolescence and impairment of equipment. During the three months ended June 30, 2024, the customer approved reimbursement of parts of the claim totaling \$4.2 million for equipment losses incurred by the Company and its vendors, which is recognized as an offset to the charge the Company recognized in the three months ended, March 31, 2024 in cost of revenues. During the three months ended September 30, 2024, the customer approved the remaining claims related to inventory obsolescence of \$2.2 million, in addition to the contractual recoverable amount, which is recognized as an offset to the charge recognized in the three months ended, March 31, 2024 in cost of revenues.

Federal, State and Local Environmental Regulations

The Company is subject to federal, state and local environmental laws and regulations. These laws generally provide for control of pollutants released into the environment and require responsible parties to undertake remediation. Penalties may be imposed for noncompliance.

Litigation

The Company is, from time to time, a party to litigation that arises in the normal course of its business operations. See Part II, Item 1 “Legal Proceedings” of this Quarterly Report on Form 10-Q for a description of certain of the Company’s current legal proceedings. The Company is not presently a party to any litigation for which it believes a loss is probable requiring an amount to be accrued or a possible loss contingency requiring disclosure.

Purchase Commitments

As of September 30, 2024, the Company had purchase commitments of approximately \$240.0 million, which included capital commitments of \$175.3 million. Purchase commitments related to capital expenditures are anticipated to be spent over the next three years, while the Company's remaining purchase commitments are anticipated to be spent throughout 2024.

Purchase obligations relate primarily to open purchase orders for capital expenditures, inventories, and goods and services. Purchase obligations are entered into with various vendors in the normal course of business and are consistent with the Company's expected requirements.

Warranty

The Company offers warranties to its customers depending upon the specific product.

The Company's standard warranty period for energy industrial products extends to one year from the date of shipment. This standard warranty provides that the Company's products will be free from defects in material and workmanship, and will, under normal use, conform to the specifications for the product.

The Company's thermal barrier products provide quality and warranty provisions customary in the automotive industry.

The Company recorded warranty reserves related to its thermal barrier products of \$1.1 million during the nine months ended September 30, 2024 and \$0.2 million during the nine months ended September 30, 2023.

(11) Leases and sale and leaseback

The Company leases office, laboratory, warehouse and fabrication space in Massachusetts, Rhode Island and Monterrey, Mexico under operating leases. Under these agreements, the Company is obligated to pay annual rent, real estate taxes, and certain other operating expenses. The Company also leases equipment under operating leases. The Company's operating leases expire at various dates through 2034.

The Company determines if an arrangement is a lease at inception. Right-of-use (ROU) assets represent the Company's right to use an underlying asset for the lease term and lease liabilities represent the Company's payment obligations under the lease. Operating lease ROU assets and liabilities are recognized based on the present value of lease payments over the lease term. To measure its lease liabilities, the Company uses its incremental borrowing rate or the rate implicit in the lease, if available. The Company calculates its incremental borrowing rate using a synthetic credit rating analysis based on Moody's Building Materials Industry Rating Methodology. ROU assets also include any direct costs and prepaid lease payments but exclude any lease incentives received. The Company's lease terms may include options to extend or terminate the lease when it is reasonably certain that the Company will exercise that option. Lease expense for lease payments is recognized on a straight-line basis over the lease term.

The Company elected the short-term lease recognition exemption for all leases that qualify. For leases that qualify for this exemption, the Company does not recognize ROU assets or lease liabilities. For lease agreements with lease and non-lease components, the Company accounts for each component separately. However, in the case of equipment leases, the Company accounts for lease and non-lease components as a single component.

Maturities of operating lease liabilities as of September 30, 2024 are as follows:

Year	Operating Leases (In thousands)
2024 (excluding the nine months ended September 30, 2024)	\$ 1,523
2025	6,110
2026	5,598
2027	4,695
2028	4,599
Thereafter	19,338
Total lease payments	41,863
Less imputed interest	(14,940)
Total lease liabilities	\$ 26,923

The Company incurred operating lease costs of \$4.3 million and \$4.2 million during the nine months ended September 30, 2024 and 2023, respectively. Cash payments related to operating lease liabilities were \$4.0 million and \$3.6 million during the nine months ended September 30, 2024 and 2023, respectively.

As of September 30, 2024, the weighted average remaining lease term for operating leases was 7.8 years. As of September 30, 2024, the weighted average discount rate for operating leases was 12.0%.

As of September 30, 2024, the Company had an additional operating equipment lease that will commence during 2024 with total lease payments of \$0.3 million and a weighted average lease term of 5.0 years.

As of September 30, 2024, the Company had no additional operating real estate leases that would commence during 2024.

Sale and leaseback transaction

In January 2024, the Company entered into a sale and leaseback arrangement, pursuant to which the Company sold certain equipment to an equipment leasing company for a one-time cash payment of \$5.0 million and leased back such equipment from the leasing company. The transaction was considered as a failed sale and leaseback transaction and accordingly, was accounted as a financing transaction. The Company did not recognize a gain on any of the proceeds received from the lessor that contractually constitute payments to acquire the assets subject to these arrangements. Instead, the sale proceeds received were accounted for as finance obligations. The monthly lease rents will be paid over the term of three years and will be allocated between interest expense and principal repayment of the financial liability.

In September 2024, the Company entered into a sale and leaseback arrangement, pursuant to which the Company sold certain equipment to an equipment leasing company for a one-time cash payment of \$10.0 million and leased back such equipment from the leasing company. The transaction was considered as a failed sale and leaseback transaction and accordingly, was accounted as a financing transaction. The Company did not recognize a gain on any of the proceeds received from the lessor that contractually constitute payments to acquire the assets subject to these arrangements. Instead, the sale proceeds received were accounted for as finance obligations. The monthly lease rents will be paid over the term of three years and will be allocated between interest expense and principal repayment of the financial liability.

The outstanding finance obligation balance as of September 30, 2024 was \$14.1 million

(12) Stock based compensation

During the nine months ended September 30, 2024, the Company granted 242,279 restricted common stock units (RSUs) with an aggregate grant date fair value of \$4.0 million and non-qualified stock options (NSOs) to purchase 569,301 shares of common stock with an aggregate grant date fair value of \$6.4 million to employees under its equity incentive plans. The restricted common stock and NSOs granted to employees will typically vest over a three-year period.

During the nine months ended September 30, 2024, the Company also granted 13,264 shares of restricted common stock with a grant date fair value of \$0.4 million and NSOs to purchase 11,618 shares of common stock with a grant date fair value of \$0.2 million to its non-employee directors under the 2023 Equity Plan. The RSUs and NSOs granted to non-employee directors will typically vest over a one-year period.

Stock-based compensation is included in cost of revenue or operating expenses, as applicable, and consists of the following:

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2024	2023	2024	2023
	(In thousands)		(In thousands)	
Cost of product revenue	\$ 186	\$ 93	\$ 545	\$ 420
Research and development expenses	188	255	891	510
Sales and marketing expenses	456	378	1,241	1,110
General and administrative expenses	1,800	2,063	7,630	5,726
Total stock-based compensation	<u>\$ 2,630</u>	<u>\$ 2,789</u>	<u>\$ 10,307</u>	<u>\$ 7,766</u>

The 2023 Equity Plan was approved by stockholders at the Company's annual meeting of stockholders on June 1, 2023 as the successor to the Company's 2014 Employee, Director and Consultant Equity Incentive Plan (the 2014 Equity Plan), and no further awards may be made under the 2014 Equity Plan after that date. As of September 30, 2024, 5,012,394 shares of common stock were reserved for issuance upon the exercise or vesting of outstanding stock-based awards granted under the Company's equity incentive plans. Any cancellations or forfeitures of awards outstanding under the 2023 Equity Plan, the 2014 Equity Plan or the 2001 Equity Incentive Plan, as amended (the 2001 Equity Plan) will result in the shares reserved for issuance pursuant to such awards becoming available for grant under the 2023 Equity Plan. As of September 30, 2024, the Company has either reserved in connection with statutory tax withholdings or issued a total of 6,609,298 shares under the Company's equity incentive plans. As of September 30, 2024, there were 2,180,098 shares of common stock available for future grant under the 2023 Equity Plan.

On March 5, 2024, the Compensation and Leadership Development Committee (the Committee) of the Board of Directors of the Company approved the cancellation of the outstanding, unearned portion of the performance-based restricted shares granted to certain employees pursuant to the 2014 Equity Plan on June 29, 2021 (to Donald R. Young) and June 2, 2022 (to certain other employees). The Committee determined that based on current market conditions, the likelihood of achievement of any of the remaining performance hurdles applicable to the unearned restricted shares is remote, and that the unearned restricted shares therefore had ceased to have incentive value for the grantees. On March 6, 2024, the Company entered into cancellation agreements, pursuant to which the applicable employees agreed to such cancellation.

The cancelled unearned restricted shares were added to the number of shares available for awards under the Company's 2023 Equity Incentive Plan. For financial accounting purposes, the cancellation of the unearned restricted shares resulted in the immediate charge of approximately \$2.2 million of unamortized stock compensation costs of which \$2.0 million is included in the general and administrative expenses and \$0.2 million is included in research and development expenses in the accompanying consolidated statement of operations.

(13) Net Income (Loss) Per Share

The computation of basic and diluted net income (loss) per share consists of the following:

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2024	2023	2024	2023
(In thousands, except share and per share data)				
Numerator:				
Net income (loss)	\$ (12,970)	\$ (13,073)	\$ 2,013	\$ (45,292)
Denominator:				
Weighted average shares outstanding, basic	76,261,294	69,317,805	76,402,123	69,243,843
Weighted average shares outstanding, diluted	76,261,294	69,317,805	79,149,193	69,243,843
Net income (loss) per share, basic	\$ (0.17)	\$ (0.19)	\$ 0.03	\$ (0.65)
Net income (loss) per share, diluted	\$ (0.17)	\$ (0.19)	\$ 0.03	\$ (0.65)

Potentially dilutive common shares that were excluded from the computation of diluted net (loss) income per share because they were anti-dilutive consist of the following:

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2024	2023	2024	2023
Common stock options	4,453,006	5,462,015	110,900	5,462,015
Restricted common stock units	559,389	568,469	1,957	568,469
Restricted common stock awards	13,264	889,366	824	889,366
Convertible note, if converted	—	3,862,221	—	3,862,221
Total	5,025,659	10,782,071	113,681	10,782,071

The potential dilutive shares from common stock options, restricted common stock units, restricted common stock awards, and the convertible note were excluded from the calculation of diluted net income (loss) per share because their effect would have been

anti-dilutive for the periods presented. The Company excludes the shares issued in connection with restricted stock awards from the calculation of basic weighted average common shares outstanding until the restrictions lapse.

(14) Income Taxes

The Company incurred net operating loss and income for the three and nine months ended September 30, 2024, respectively. The Company incurred net operating losses and recorded a full valuation allowance against net deferred tax assets for all prior periods. Accordingly, the Company has not recorded a provision for federal or state income taxes for the three and nine months ended September 30, 2024. The Company has provided \$0.3 million of income tax expense related to its Mexican maquiladora operations for the three months ended September 30, 2024. The Company has provided \$1.9 million of income tax expense related to its Mexican maquiladora operations for the nine months ended September 30, 2024.

(15) Segment Information

Operating segments are identified as components of an enterprise about which separate, discrete financial information is available for evaluation by the chief operating decision maker in making decisions on how to allocate resources and assess performance. The Company's chief operating decision maker is the Chief Executive Officer. The Company's chief operating decision maker reviews consolidated operating results to make decisions about allocating resources and assessing performance for the entire Company. The Company reports two segments: Energy Industrial and Thermal Barrier. The Company evaluates segment performance based on the segment profit (loss) before corporate expenses.

Summarized below are the Revenue and Segment Operating Profit for each reporting segment:

	Revenue Three Months Ended September 30,		Segment Operating Profit (Loss) Three Months Ended September 30,		Revenue Nine Months Ended September 30,		Segment Operating Profit (Loss) Nine Months Ended September 30,	
	2024	2023	2024	2023	2024	2023	2024	2023
	(In thousands)		(In thousands)		(In thousands)		(In thousands)	
Energy industrial	\$ 26,776	\$ 27,912	\$ 10,733	\$ 5,825	\$ 92,781	\$ 97,311	\$ 37,750	\$ 24,246
Thermal barrier	90,564	32,843	38,310	7,985	236,830	57,188	98,014	3,057
Total	<u>\$ 117,340</u>	<u>\$ 60,755</u>	<u>\$ 49,043</u>	<u>\$ 13,810</u>	<u>\$ 329,611</u>	<u>\$ 154,499</u>	<u>\$ 135,764</u>	<u>\$ 27,303</u>
Corporate expenses			31,643	28,444			95,942	77,889
Operating profit (loss)			17,400	(14,634)			39,822	(50,586)
Other (expense) income, net			(30,103)	1,561			(35,920)	5,294
Income tax expense			(267)	-			(1,889)	-
Net income (loss)			<u>\$ (12,970)</u>	<u>\$ (13,073)</u>			<u>\$ 2,013</u>	<u>\$ (45,292)</u>

	Total Assets	
	September 30, 2024	December 31, 2023
	(In thousands)	
Energy industrial	\$ 97,575	\$ 93,168
Thermal barrier	172,582	118,565
Total assets of reportable segments	270,157	211,733
Construction in progress	344,041	314,678
All other corporate assets	168,389	176,637
	<u>\$ 782,587</u>	<u>\$ 703,048</u>

(16) Subsequent Events

The Company has evaluated subsequent events through November 7, 2024, the date of issuance of the consolidated financial statements for the three and nine months ended September 30, 2024.

In October 2024, the Company entered into an underwriting agreement (the Underwriting Agreement) with Goldman Sachs & Co. LLC and Morgan Stanley & Co. LLC (collectively, the Underwriters), pursuant to which the Company issued and sold an aggregate of 4,887,500 shares of the Company's common stock, which included 637,500 shares pursuant to the Underwriters' option to purchase additional shares of common stock of the Company, to the Underwriters in a registered underwritten offering (the "Offering"). The price to the public in the Offering was \$20.00 per share. The net proceeds to the Company from the Offering were approximately \$93.2 million, after deducting underwriting discounts and commissions and estimated offering expenses payable by the Company.

Item 2. Management's Discussion and Analysis of Financial Condition and Results of Operations.

The following information should be read in conjunction with the unaudited financial information and the notes thereto included in this Quarterly Report on Form 10-Q and the audited financial information and the notes thereto included in the Annual Report on Form 10-K for the year ended December 31, 2023, filed with the U.S. Securities and Exchange Commission (SEC) on March 7, 2024, which we refer to as the Annual Report.

Certain matters discussed in this Quarterly Report on Form 10-Q may be deemed to be forward-looking statements that involve risks and uncertainties. We make such forward-looking statements pursuant to the safe harbor provisions of the Private Securities Litigation Reform Act of 1995 and other federal securities laws. In this Quarterly Report on Form 10-Q, words such as "may," "will," "anticipate," "estimate," "expects," "projects," "intends," "plans," "believes" and similar expressions (as well as other words or expressions referencing future events, conditions or circumstances) are intended to identify forward-looking statements.

Our actual results and the timing of certain events may differ materially from the results discussed, projected, anticipated, or indicated in any forward-looking statements. We caution you that forward-looking statements are not guarantees of future performance and that our actual results of operations, financial condition and liquidity, and the development of the industry in which we operate may differ materially from the forward-looking statements contained in this Quarterly Report on Form 10-Q. In addition, even if our results of operations, financial condition and liquidity, and the development of the industry in which we operate are consistent with the forward-looking statements contained in this Quarterly Report on Form 10-Q, they may not be predictive of results or developments in future periods.

The following information and any forward-looking statements should be considered in light of factors discussed elsewhere in this Quarterly Report on Form 10-Q and under "Risk Factors" in Item 1A of the Annual Report.

We caution readers not to place undue reliance on any forward-looking statements made by us, which speak only as of the date they are made. We disclaim any obligation, except as specifically required by law and the rules of the SEC, to publicly update or revise any such statements to reflect any change in our expectations or in events, conditions or circumstances on which any such statements may be based, or that may affect the likelihood that actual results will differ from those set forth in the forward-looking statements.

You should read the following discussion and analysis of financial condition and results of operations together with Part I Item 1 "Financial Statements," which includes our financial statements and related notes, elsewhere in this Quarterly Report on Form 10-Q.

Investors and others should note that we routinely use the Investors section of our website to announce material information to investors and the marketplace. While not all of the information that we post on the Investors section of our website is of a material nature, some information could be deemed to be material. Accordingly, we encourage investors, the media, and others interested in us to review the information that we share on the Investors section of our website, <https://www.aerogel.com>.

Products

Our core businesses are organized into two reportable segments: Energy Industrial and Thermal Barrier. The following describes our key product offerings and new product innovations by reportable segment.

Thermal Barrier

We are actively developing a number of promising aerogel products and technologies for the electric vehicle (EV) market. We have developed and are commercializing our proprietary line of PyroThin[®] aerogel thermal barriers for use in battery packs in EVs. Our PyroThin product is an ultra-thin, lightweight and flexible thermal barrier designed with other functional layers to impede the propagation of thermal runaway across multiple lithium-ion battery system architectures. Our thermal barrier technology is designed to offer a unique combination of thermal management, mechanical performance and fire protection properties. These properties enable EV manufacturers to achieve critical battery performance and safety goals. In addition, we are seeking to leverage our patented carbon aerogel technology to develop industry-leading battery materials for use in lithium-ion battery cells. These battery materials have the potential to increase the energy density of the battery cells, thus enabling an increase in the driving range of EVs.

The commercial potential for our PyroThin thermal barriers and our carbon aerogel battery materials in the EV market is significant. Accordingly, we are hiring additional personnel, incurring additional operating expenses, incurring significant capital expenditures to expand aerogel manufacturing capacity, establishing an automated thermal barrier fabrication operation, enhancing research and development resources and expanding our battery material research facilities, among other items.

We have entered into production contracts with certain major OEMs, including General Motors LLC (GM), to supply fabricated, multi-part thermal barriers for use in the battery system of its next-generation EVs. Pursuant to the contracts with GM, we are obligated to supply the barriers at fixed annual prices and at volumes to be specified by the customer up to a daily maximum quantity through the term of the agreements, which expire at various times from 2026 through 2034. While GM has agreed to purchase its requirement for the barriers from us at locations to be designated from time to time, it has no obligation to purchase any minimum quantity of barriers under the contracts. In addition, GM may terminate the contracts any time and for any or no reason. All other terms of the contracts are generally consistent with GM's standard purchase terms, including quality and warranty provisions customary in the automotive industry. We have also entered into production contracts with Toyota, Scania, Audi, a luxury brand of the Volkswagen Group, ACC, a battery cell joint venture between Stellantis N.V, Saft-TotalEnergies and Mercedes-Benz, and a large EU battery manufacturer to supply a next generation vehicle platform of a major EU luxury sports car brand.

Energy Industrial

We also design, develop and manufacture innovative, high-performance aerogel insulation used primarily in the energy industrial and sustainable insulation materials markets. We believe our aerogel blankets deliver the best thermal performance of any widely used insulation product available on the market today and provide a combination of performance attributes unmatched by traditional insulation materials. Our end-user customers select our products where thermal performance is critical and to save money, improve resource efficiency, enhance sustainability, preserve operating assets and protect workers. Our insulation is used by oil producers and the owners and operators of refineries, petrochemical plants, LNG facilities, power generating assets and other energy industrial companies. Our Pyrogel® and Cryogel® product lines have undergone rigorous technical validation by industry leading end-users and achieved significant market adoption.

We also derive revenue from a number of other end markets. Customers in these markets use our products for applications as diverse as military and commercial aircraft, trains, buses, appliances, apparel, footwear and outdoor gear. As we continue to enhance our Aerogel Technology Platform®, we believe we will have additional opportunities to address high-value applications in the global insulation market, and in a number of new, high-value markets, including hydrogen energy, filtration, water purification, and gas sorption.

We market and sell our products primarily through a sales force based in North America, Europe and Asia. The efforts of our sales force are supported by a small number of sales consultants with extensive knowledge of a particular market or region. Our sales force is responsible for establishing and maintaining customer and partner relationships, delivering highly technical information and ensuring high-quality customer service.

Our salespeople work directly with end-user customers and engineering firms to promote the qualification, specification and acceptance of our aerogel and thermal barrier products. We also rely on an existing and well-established channel of qualified insulation distributors and contractors in more than 50 countries around the world to ensure rapid delivery of our aerogel products and strong end-user support.

Manufacturing Operations

We manufacture our products using our proprietary technology at our facility in East Providence, Rhode Island. We have operated the East Providence facility since 2008 and have increased our capacity in phases. To meet expected growth in demand for our aerogel products in the EV market, we have been in the process of expanding our aerogel blanket capacity by constructing a second manufacturing plant in Bulloch County, Georgia. However, in order to manage the development of the second plant so that its increased capacity comes online in a manner that aligns with our current expectations as to demand from our EV customers, we are extending the timeframe for construction and commissioning of the second plant until such time as its capacity is supported by increased demand. In the meantime, and until we ramp up construction, we believe that productivity improvements in our existing Rhode Island facility combined with our supplemental supply of our energy industrial products from an external manufacturing facility in China will permit us to achieve our target revenue capacity in 2024 and prior to the completion and start-up of the second plant. At present, we have engaged one third-party external manufacturing facility in China to produce certain of our Energy Industrial products. Pursuant to our supply contract with this contract manufacturer, they are obligated to deliver products to us as we issue purchase orders on an as-needed basis through the term of the agreement. The contract automatically renews year-to-year unless either party notifies the other of its intention not to renew the contract. While we have agreed to purchase our requirement for certain Energy Industrial products from the contract manufacturer, we have no obligation to purchase any minimum quantity under the contract and we may terminate the contract at any time and for any or no reason. Additionally, we have entered into a contract with Prodensa Servicios de Consultora (Prodensa) to establish OPE Manufacturer Mexico S de RL de CV, a maquiladora located in Mexico, (OPE) which manufactures thermal barrier PyroThin products and operates an automated fabrication facility for PyroThin. Pursuant to such contract, we pay Prodensa a management fee and have an option to purchase OPE from Prodensa after a period of 18 months.

Recent Developments

Underwritten Offering

In October 2024, we entered into an underwriting agreement (the Underwriting Agreement) with Goldman Sachs & Co. LLC and Morgan Stanley & Co. LLC (collectively, the Underwriters), pursuant to which we issued and sold an aggregate of 4,887,500 shares of our common stock, which included 637,500 shares pursuant to the Underwriters' option to purchase additional shares of our common stock, to the Underwriters in a registered underwritten offering (the "Offering"). The price to the public in the Offering was \$20.00 per share. The net proceeds to us from the Offering were approximately \$93.2 million, after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

U.S. Department of Energy Loan Programs Office

On October 16, 2024, we announced that the U.S. Department of Energy (DOE) Loan Programs Office provided us a conditional commitment for a proposed loan of up to \$670.6 million (the Loan) under the Advanced Technology Vehicles Manufacturing (ATVM) loan program within DOE's Loan Programs Office (LPO) for financing the construction of our planned second aerogel manufacturing facility (the Statesboro Plant) in Statesboro, Georgia. Aspen Aerogels Georgia, LLC, a subsidiary of Aspen Aerogels, Inc, is the intended borrower of the Loan. The Loan would be structured as project financing, providing for repayment commencing upon project commissioning. The interest rate will be the applicable U.S. Treasury Rate at the time of each cash draw. While this conditional commitment indicates the DOE's intent to finance the project, the DOE and the Company must satisfy certain technical, legal, environmental, and financial conditions before the DOE enters into definitive financing documents and funds the loan.

We had previously estimated the cost of the Statesboro Plant to be approximately \$710.0 million. Adjusting the estimated cost to account for the right-timing of construction, inflation, and a revised scope that reflects recent process improvements, we have revised our aggregate cost estimate to a range of approximately \$800.0 million to \$960.0 million. The prospective DOE loan is intended to fully fund our remaining spend on the construction of the plant, following and assuming the expected closing of the Loan.

The receipt by us of the conditional commitment from the DOE in respect of the Loan is not an assurance that definitive financing documents will be negotiated and entered into, that the terms and conditions of the Loan will be consistent with the terms contemplated in the conditional commitment, or that all technical, legal, environmental and financial conditions will be achieved and satisfied. The outcome of the process with the DOE relating to the Loan and the Company's ability to draw and utilize such funds towards the construction of the Statesboro Plant is dependent on whether the DOE determines to proceed with the definitive financing documents and under what conditions. There can be no assurance as to the outcome of these decisions and the terms and conditions of the financing documents that may be offered, if any. See the section titled "Risk Factors—Risks Related to Our Business and Strategy—We are dependent upon our ability to close on and fully draw down on our proposed loan facility from the United States Department of Energy, which may restrict our ability to conduct our business."

MidCap Loan Facility

On August 19, 2024, we and Aspen Aerogels Rhode Island, LLC, a Rhode Island limited liability company (Aspen RI and, together with the Company, each, a Borrower and collectively, the "Borrowers") entered into a Credit, Security and Guaranty Agreement (the Credit Agreement and the facilities provided thereunder, collectively, the MidCap Loan Facility), by and among the Borrowers, MidCap Funding IV Trust, as agent (the Agent), MidCap Financial Trust, as term loan servicer, the financial institutions or other entities from time to time party thereto as lenders (the Lenders), and the other parties party thereto as additional guarantors and/or borrowers from time to time. The proceeds of the MidCap Loan Facility were used in connection with the transaction contemplated by the Note Repurchase Agreement (as defined below), the payment of related fees and expenses and for working capital of the Company and its subsidiaries. Loans borrowed under the MidCap Loan Facility mature on August 19, 2029.

The MidCap Loan Facility is comprised of (i) a term loan facility in an aggregate principal amount of \$125.0 million (the Term Loan Facility) and (ii) an asset-based revolving credit facility in an aggregate principal amount not to exceed the lesser of (A) a \$100.0 million commitment amount and (B) the value of the borrowing base (defined as the sum of (x) 85% of certain eligible accounts of the Borrowers and (y) the lesser of 85% of the NOLV or 85% of the cost of certain eligible inventory of the Borrowers) (the Revolving Facility). Loans borrowed under the Term Loan Facility will bear an interest rate equal to Term SOFR (as defined in the Credit Agreement) for a one-month interest period plus 4.50% per year, subject to a Term SOFR floor of 4.50% and a Term SOFR cap of 7.50%. Loans borrowed under the Revolving Facility will bear an interest rate equal to Term SOFR plus 4.60% per year, subject to a Term SOFR floor of 2.50%. The Term Loan Facility is subject to amortization of principal, payable quarterly on the last day of each quarter, commencing September 30, 2024, in an amount as set forth in the Credit Agreement with the remaining aggregate principal amount payable on the maturity date. The Revolving Facility has a required minimum balance set at 30% of the average borrowing

base during the immediate preceding month. The Borrowers are required to pay the Lenders under the Revolving Facility an unused line fee of 0.30% of the average unused availability under the Revolving Facility, subject to the aforementioned minimum balance.

Repurchase of Convertible Note

On August 19, 2024, we entered into a note purchase and sale agreement (the Note Repurchase Agreement) with Wood River Capital, LLC (Wood River), an entity affiliated with Koch Disruptive Technologies, LLC, pursuant to which we repurchased from Wood River \$123.9 million in aggregate capitalized principal amount (inclusive of PIK interest paid through June 30, 2024) of Convertible Senior PIK Toggle Notes due 2027, dated February 18, 2022, as amended by Amendment No. 1 to Convertible Senior PIK Toggle Notes due 2027, dated November 28, 2022 (the Convertible Note), such aggregate amount being the entire outstanding amount of the Convertible Note, for a total purchase price of \$150.1 million in cash, which amount equals to the Redemption Price (as defined in the Convertible Note). Pursuant to the Note Repurchase Agreement, all rights and obligations, covenants and agreements under the Convertible Note and the underlying note purchase agreement were satisfied and discharged.

Financial Summary

Our revenue for the nine months ended September 30, 2024 was \$329.6 million, which represented an increase of \$175.1 million, or 113%, from \$154.5 million for the nine months ended September 30, 2023. Net income for the nine months ended September 30, 2024 was \$2.0 million and net loss per share was \$0.03. Net loss for the nine months ended September 30, 2023 was \$45.3 million and net loss per share was \$0.65.

Key Metrics and Non-GAAP Financial Measures

We regularly review a number of metrics, including the following key metric, to evaluate our business, measure our performance, identify trends affecting our business, formulate financial projections and make strategic decisions.

Adjusted EBITDA

We use Adjusted EBITDA, a non-GAAP financial measure, as a means to assess our operating performance. We define Adjusted EBITDA as net income (loss) before interest expense, taxes, depreciation, amortization, stock-based compensation expense and other items, from time to time, which we do not believe are indicative of our core operating performance. Adjusted EBITDA is a supplemental measure of our performance that is not presented in accordance with accounting principles generally accepted in the United States of America (U.S. GAAP). Adjusted EBITDA should not be considered as an alternative to net income (loss) or any other measure of financial performance calculated and presented in accordance with U.S. GAAP. In addition, our definition and presentation of Adjusted EBITDA may not be comparable to similarly titled measures presented by other companies.

We use Adjusted EBITDA:

- as a measure of operating performance because it does not include the impact of items that we do not consider indicative of our core operating performance;
- for planning purposes, including the preparation of our annual operating budget;
- to allocate resources to enhance the financial performance of our business; and
- as a performance measure used under our bonus plan.

We also believe that the presentation of Adjusted EBITDA provides useful information to investors with respect to our results of operations and in assessing the performance and value of our business. Various measures of EBITDA are widely used by investors to measure a company's operating performance without regard to items that can vary substantially from company to company depending upon financing and accounting methods, book values of assets, capital structures and the methods by which assets were acquired.

Although measures similar to Adjusted EBITDA are frequently used by investors and securities analysts in their evaluation of companies, we understand that Adjusted EBITDA has limitations as an analytical tool, and you should not consider it in isolation or as a substitute for net income (loss), income (loss) from operations, net cash provided by (used in) operating activities or an analysis of our results of operations as reported under U.S. GAAP. Some of these limitations are:

- Adjusted EBITDA does not reflect our historical cash expenditures or future requirements for capital expenditures or other contractual commitments;

- Adjusted EBITDA does not reflect changes in, or cash requirements for, our working capital needs;
- Adjusted EBITDA does not reflect stock-based compensation expense;
- Adjusted EBITDA does not reflect our income tax expense or cash requirements to pay our income taxes;
- Adjusted EBITDA does not reflect our interest expense, or the cash requirements necessary to service interest or principal payments on our debt;
- although depreciation, amortization and impairment charges are non-cash charges, the assets being depreciated, amortized or impaired will often have to be replaced in the future, and Adjusted EBITDA does not reflect any cash requirements for these replacements; and
- other companies in our industry may calculate EBITDA or Adjusted EBITDA differently than we do, limiting their usefulness as a comparative measure.

Because of these limitations, our Adjusted EBITDA should not be considered as a measure of discretionary cash available to us to reinvest in the growth of our business or as a measure of cash available for us to meet our obligations.

To properly and prudently evaluate our business, we encourage you to review the U.S. GAAP financial statements included elsewhere in this Quarterly Report on Form 10-Q, and not to rely on any single financial measure to evaluate our business.

The following table presents a reconciliation of net income (loss), the most directly comparable U.S. GAAP measure, to Adjusted EBITDA for the periods presented:

	Three Months Ended September 30,		Nine Months Ended September 30,	
	2024	2023	2024	2023
	(In thousands)			
Net income (loss)	\$ (12,970)	\$ (13,073)	\$ 2,013	\$ (45,292)
Depreciation and amortization	5,321	4,550	17,093	10,757
Stock-based compensation ⁽¹⁾	2,630	2,789	10,307	7,766
Other expense (income)	2,616	(1,561)	8,433	(5,294)
Loss on extinguishment of debt	27,487	-	27,487	-
Income tax expense	267	-	1,889	-
Adjusted EBITDA	<u>\$ 25,351</u>	<u>\$ (7,295)</u>	<u>\$ 67,222</u>	<u>\$ (32,063)</u>

⁽¹⁾ Represents non-cash stock-based compensation related to vesting and modifications of stock option grants, vesting of restricted stock units and vesting of restricted common stock.

Our financial performance, including such measures as net income (loss), earnings per share and Adjusted EBITDA, are affected by a number of factors including volume and mix of aerogel products sold, average selling prices, our material costs and manufacturing expenses, the costs associated with capacity expansions and start-up of additional production capacity, and the amount and timing of operating expenses. Accordingly, we expect that our net income (loss), earnings per share and Adjusted EBITDA will vary from period to period.

We expect to maintain strong revenue growth during 2024 driven by accelerating demand in the EV market and continued market share gains in the sustainable insulation materials market. Our expectation to maintain strong revenue growth is based, in part, on our OEM customers' production volume forecasts and targets as well as our expectation to successfully scale our manufacturing capabilities and address any potential supply chain issues to meet this expected demand. As a result, we expect to experience a decrease in both net loss and negative Adjusted EBITDA during 2024, compared to 2023.

Components of Our Results of Operations

Revenue

We recognize revenue from the sale of our energy industrial aerogel products and thermal barriers. Revenue is recognized upon the satisfaction of contractual performance obligations.

We record deferred revenue for sales when (i) we have delivered products, but other revenue recognition criteria have not been satisfied, or (ii) payments have been received in advance of the completion of required performance obligations.

We project revenue growth during 2024 due to accelerating demand in the EV market and continued market share gains in the sustainable insulation materials market.

Cost of Revenue

Cost of product revenue consists primarily of materials and manufacturing expense. Cost of product revenue is recorded when the related product revenue is recognized.

Material is a significant component of cost of product revenue and includes fibrous batting, silica materials and additives. Material costs as a percentage of product revenue vary from product to product due to differences in average selling prices, material requirements, product thicknesses, and manufacturing yields. In addition, we provide warranties for our products and record the estimated cost within cost of revenue in the period that the related revenue is recorded or when we become aware that a potential warranty claim is probable and can be reasonably estimated. As a result of these factors, material costs as a percentage of product revenue will vary from period to period due to changes in the mix of aerogel products sold, the costs of our raw materials or the estimated cost of warranties. In addition, global supply chain disturbances, increased reliance on foreign materials procurement, industrial gas supply constraints, increases in the cost of our raw materials, engineering changes, higher prototype sales and other factors may significantly impact our material costs and have a material impact on our operations. We expect that material costs will increase in absolute dollars during 2024 due to projected growth in product shipments and contracts but remain stable as a percentage of revenue due to improved manufacturing, and fabrication yields and a favorable mix of products sold.

Manufacturing expense is also a significant component of cost of revenue. Manufacturing expense includes labor, utilities, maintenance expense, and depreciation on manufacturing assets. Manufacturing expense also includes stock-based compensation of manufacturing employees and shipping costs. We expect that manufacturing expense will increase in absolute dollars and decrease as a percentage of revenue during 2024 due to increased staffing costs and spending levels in support of our thermal barrier business, including the operation of an automated fabrication facility in Monterrey, Mexico.

We are also continuing to monitor the impact on our material costs, manufacturing expense, and cost of product revenue from engaging one or more external manufacturing facilities in China to supply our aerogel products for the energy industrial market beginning in 2024. We expect that our energy industrial product material costs will increase in absolute dollars and as a percentage of revenue, and manufacturing expense and cost of revenue will decrease as a percentage of revenue due to projected volume growth, which will be manufactured through the external manufacturing facility.

During 2024, we expect that cost of product revenue will increase in absolute dollars due to projected volume growth and a planned increase in staffing and spending levels, but decrease as a percentage of product revenue due to projected increases in average selling prices, improved manufacturing and fabrication yields and a favorable mix of products sold.

Gross Profit

Our gross profit as a percentage of revenue is affected by a number of factors, including the volume of products produced and sold, the mix of products sold, average selling prices, our material and manufacturing costs, realized capacity utilization and the costs associated with expansions and start-up of production capacity. Accordingly, we expect our gross profit to vary significantly in absolute dollars and as a percentage of revenue from period to period.

During 2024, we expect gross profit to increase in both absolute dollars and as a percentage of total revenue due to the combination of a projected increase in total revenue combined with projected reduction in material costs and manufacturing expense as a percentage of total revenue.

In the longer term, we expect gross profit to improve in absolute dollars and as a percentage of revenue due to expected increases in total revenue, production volumes and manufacturing productivity. In addition, we expect the gross profit improvement derived from the increases in revenue, volume and productivity will be supported by the continued implementation of lower cost product formulations and realization of material purchasing efficiencies.

Operating Expenses

Operating expenses consist of research and development, sales and marketing, and general and administrative expenses. Operating expenses include personnel costs, legal fees, professional fees, service fees, insurance premiums, travel expense, facilities related costs and other costs, expenses and fees. The largest component of our operating expenses is personnel costs, consisting of

salaries, benefits, incentive compensation and stock-based compensation. In any particular period, the timing and extent of personnel additions or reductions, legal activities, including patent enforcement actions, marketing programs, research efforts and a range of similar activities or actions could materially affect our operating expenses, both in absolute dollars and as a percentage of revenue.

Research and Development Expenses

Research and development expenses consist primarily of expenses for personnel engaged in the development of next generation aerogel compositions, form factors and manufacturing technologies. These expenses also include testing services, prototype expenses, consulting services, trial formulations for new products, equipment depreciation, facilities costs and related overhead. We expense research and development costs as incurred. We expect to continue to devote substantial resources to the development of new aerogel technologies, including our carbon aerogel battery materials. We believe that these investments are necessary to maintain and improve our competitive position. We also expect to continue to invest in research and engineering personnel and the infrastructure required in support of their efforts. We expect our research and development expenses will increase in absolute dollars and decrease as a percentage of revenue in 2024 and the longer term.

Sales and Marketing Expenses

Sales and marketing expenses consist primarily of personnel costs, incentive compensation, marketing programs, costs of new product and process introductions, travel and related costs, consulting expenses and facilities related costs. We expect our sales and marketing expenses will increase in absolute dollars but decrease as a percentage of revenue in 2024 and in the longer term.

General and Administrative Expenses

General and administrative expenses consist primarily of personnel costs, legal expenses, consulting and professional services, audit fees, compliance with securities, corporate governance and related laws and regulations, investor relations and insurance premiums, including director and officer insurance.

We expect our general and administrative expenses to increase as we add general and administrative personnel to support the anticipated growth of our business. We also expect that the patent enforcement actions, described in more detail under “Legal Proceedings” in Part I, Item 3 of our Annual Report and “Legal Proceedings” in Part II, Item 1 of this Quarterly Report on Form 10-Q, if protracted, could result in significant legal expense over the medium to long-term. We expect that our general and administrative expenses will increase in absolute dollars and decrease as a percentage of revenue.

Interest Expense, Convertible Note - Related Party

Interest expense, convertible note - related party is net of the capitalized interest related to the \$100.0 million in aggregate principal amount of our Convertible Senior PIK Toggle Notes due 2027, which we sold and issued to Wood River.

Interest Income (Expense)

Interest expense consists of interest expense and amortization or write-off of deferred financing costs related to our other financing arrangements including a failed sale and leaseback arrangement accounted as a financing transaction and interest earned on the cash balances invested in deposit accounts, money market accounts, and high-quality debt securities issued by the U.S. government.

Provision for Income Taxes

We have incurred annual net losses since inception and have not recorded benefit provisions for U.S. federal income taxes or state income taxes since the tax benefits of our net losses have been offset by valuation allowances due to the uncertainty associated with the utilization of net operating loss carryforwards. We record tax expenses in connection with our Mexican maquiladora operations.

Results of Operations

Three months ended September 30, 2024 compared to the three months ended September 30, 2023

The following tables set forth a comparison of the components of our results of operations for the periods presented:

Revenue

	Three Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Revenue:						
Energy industrial	\$ 26,776	23%	\$ 27,912	46%	\$ (1,136)	(4)%
Thermal barrier	90,564	77%	32,843	54%	57,721	176%
Total revenue	\$ 117,340	100%	\$ 60,755	100%	\$ 56,585	93%

Total revenue increased \$56.5 million, or 93%, to \$117.3 million for the three months ended September 30, 2024 from \$60.8 million in the comparable period in 2023. The increase in total revenue was the result of an increase in thermal barrier revenue, offset by a decrease in energy industrial revenue.

Energy industrial revenue decreased by \$1.2 million, or 4%, to \$26.7 million for the three months ended September 30, 2024 from \$27.9 million in the comparable period in 2023. This decrease was driven by a decrease in project-based demand in the subsea market and the global petrochemical and refinery markets of Asia, offset in part by an increase in the global petrochemical and refinery markets of Latin America, Europe, and North America.

Energy industrial revenue for the three months ended September 30, 2024 included \$5.6 million to a North American distributor, in comparison to \$8.0 million for the comparable period of 2023.

The average selling price per square foot of our energy industrial products increased by 6% for the three months ended September 30, 2024, compared to the three months ended September 30, 2023. The increase in average selling price reflected the impact of price increases enacted in 2024 and a change in the mix of products sold, as we strive to maximize capacity in our aerogel manufacturing facility. This increase in average selling price had the effect of increasing product revenue by \$1.4 million for the three months ended September 30, 2024 from the comparable period in 2023.

In volume terms, energy industrial product shipments decreased by 9% as measured by square feet of our energy industrial products shipped for the three months ended September 30, 2024, compared to the three months ended September 30, 2023. The decrease in volume had the effect of increasing product revenue by \$2.5 million for the three months ended September 30, 2024 from the comparable period in 2023.

Thermal barrier revenue was \$90.5 million for the three months ended September 30, 2024 as compared to \$32.8 million for the three months ended September 30, 2023. During the three months ended September 30, 2024 and 2023, thermal barrier revenue included \$86.4 million and \$29.5 million, respectively, to a major U.S. automotive OEM.

Cost of Revenue

	Three Months Ended September 30,					
	2024		2023		Change	
	Amount	Percentage of Related Revenue	Amount	Percentage of Related Revenue	Amount	Percentage
(\$ in thousands)						
Cost of revenue:						
Energy industrial	\$ 16,043	60%	\$ 22,087	79%	\$ (6,044)	(27)%
Thermal barrier	52,254	58%	24,858	76%	27,396	110%
Total cost of revenue	<u>\$ 68,297</u>	58%	<u>\$ 46,945</u>	77%	<u>\$ 21,352</u>	45%

Total cost of revenue increased \$21.3 million, or 45%, to \$68.2 million for the three months ended September 30, 2024 from \$46.9 in the comparable period in 2023. The increase in total cost of revenue was the result of an increase in thermal barrier cost of revenue, offset by a decrease in energy industrial cost of revenue.

Energy industrial cost of revenue decreased \$6.0 million, or 27%, to \$16.0 million for the three months ended September 30, 2024 from \$22.0 million in the comparable period in 2023, primarily due to a decrease in the volume of products shipped. The \$6.0 million decrease was the result of a \$5.0 million decrease in manufacturing and other operating costs due to lower volume of Energy industrial products manufactured at our plant as we move manufacturing to the external manufacturing facility, and a \$1.0 million decrease in material costs.

Thermal barrier cost of revenue increased \$27.3 million to \$52.2 million for the three months ended September 30, 2024 as compared to \$24.9 million for the three months ended September 30, 2023. The \$27.3 million increase was the result of a \$21.4 million increase in material costs and a \$5.9 million increase in manufacturing costs, primarily driven by an increase in volume. Thermal barrier cost of revenue includes a credit of \$2.2 million for reimbursement of costs related to the impact from an engineering change notified by a customer to a part that we manufacture for that customer.

Gross Profit

	Three Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
(\$ in thousands)						
Gross profit:						
Energy industrial	\$ 10,733	40%	\$ 5,825	21%	\$ 4,908	84%
Thermal barrier	38,310	42%	7,985	24%	30,325	(380)%
Total gross profit	<u>\$ 49,043</u>	42%	<u>\$ 13,810</u>	23%	<u>\$ 35,233</u>	255%

Gross profit increased by \$35.2 million, or 255%, to \$49.0 million for the three months ended September 30, 2024 from \$13.8 million in the comparable period in 2023. The increase in gross profit was the result of the \$56.5 million increase in total revenue, offset by the \$21.3 million increase in total cost of revenue.

Research and Development Expenses

	Three Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
(\$ in thousands)						
Research and development expenses	\$ 4,591	4%	\$ 4,218	7%	\$ 373	9%

Research and development expenses increased by \$0.4 million, or 9%, to \$4.6 million for the three months ended September 30, 2024 from \$4.2 million in the comparable period in 2023. The \$0.4 million increase reflects increases in compensation and related costs of \$0.2 million, operating supplies and testing of \$0.1 million and other expenses of \$0.1 million.

Research and development expenses as a percentage of total revenue decreased to 4% of total revenue for the three months ended September 30, 2024 from 7% in the comparable period in 2023.

Sales and Marketing Expenses

	Three Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
Sales and marketing expenses	\$ 9,306	8%	\$ 8,386	14%	\$ 920	11%

Sales and marketing expenses increased by \$0.9 million, or 11%, to \$9.3 million for the three months ended September 30, 2024 from \$8.4 million in the comparable period in 2023. The \$0.9 million increase was principally the result of increases in compensation and related costs of \$0.4 million, professional fees of \$0.3 million, and utilities expenses of \$0.2 million.

Sales and marketing expenses as a percentage of total revenue decreased to 8% of total revenue for the three months ended September 30, 2024 from 14% in the comparable period in 2023.

General and Administrative Expenses

	Three Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
General and administrative expenses	\$ 17,746	15%	\$ 15,840	26%	\$ 1,906	12%

General and administrative expenses increased by \$1.9 million, or 12%, to \$17.7 million for the three months ended September 30, 2024 from \$15.8 million in the comparable period in 2023. The \$1.9 million increase was the result of increases in insurance costs of \$0.8 million, utilities expenditures of \$0.7 million, foreign currency transaction losses of \$0.6 million, and professional fees of \$0.3 million, offset in part by a decrease in compensation and related costs of \$0.3 million and other expenses of \$0.2 million.

General and administrative expenses as a percentage of total revenue decreased to 15% for the three months ended September 30, 2024 from 26% in the comparable period in 2023.

Other Income (Expense), net

	Three Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Other income (expense):						
Interest (expense), related party	\$ (1,469)	(1)%	\$ (1,938)	(3)%	\$ 469	(24)%
Interest income (expense), net	(1,147)	(1)%	1,313	2%	(2,460)	(187)%
Loss on extinguishment of debt	(27,487)	(23)%	—	—	(27,487)	NM
Income from Employee Retention Credits	—	—	2,186	4%	(2,186)	(100)%
Total other income (expense), net	\$ (30,103)	(26)%	\$ 1,561	3%	\$ (31,664)	(2028)%

Other income (expense), net decreased by \$31.7 million to \$30.1 million of other expense for the three months ended September 30, 2024 from \$1.6 million of other income (expense), net in the comparable period in 2023. The \$31.7 million decrease was the result of a \$27.5 million loss on extinguishment of debt, a \$0.9 million net impact of capitalized interest relating to our Convertible Note in the comparable period in 2023, a \$2.2 million income from Employee Retention Credits in the comparable period in 2023 not repeated in 2024, and a \$1.1 million increase of interest expense.

Income Tax Expense

	Three Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
Income tax expense	\$ 267	0%	\$ —	0%	\$ 267	NM

The \$0.3 million of income tax expense for the three months ended September 30, 2024 is related to our maquiladora operations in Mexico. We did not incur income tax expense for the comparable period in 2023.

Results of Operations

Nine months ended September 30, 2024 compared to the nine months ended September 30, 2023

The following tables set forth a comparison of the components of our results of operations for the periods presented:

Revenue

	Nine Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Revenue:						
Energy industrial	\$ 92,781	28%	\$ 97,311	63%	\$ (4,530)	(5)%
Thermal barrier	236,830	72%	57,188	37%	179,642	314%
Total revenue	\$ 329,611	100%	\$ 154,499	100%	\$ 175,112	113%

Total revenue increased \$175.1 million, or 113%, to \$329.6 million for the nine months ended September 30, 2024 from \$154.5 million in the comparable period in 2023. The increase in total revenue was the result of an increase in thermal barrier revenue, offset by a decrease energy industrial revenue.

Energy industrial revenue decreased by \$4.5 million, or 5%, to \$92.8 million for the nine months ended September 30, 2024 from \$97.3 million in the comparable period in 2023. This decrease was driven by a decrease in revenue to the global petrochemical and refinery markets of Asia and project-based demand in the subsea market, offset in part by an a more favorable mix of product shipments in the global petrochemical and refinery markets in North America, Latin America and Europe.

Energy industrial revenue for the nine months ended September 30, 2024 included \$19.1 million to a North American distributor, in comparison to \$27.6 million for the comparable period of 2023.

The average selling price per square foot of our energy industrial products increased by 12% for nine months ended September 30, 2024, compared to the nine months ended September 30, 2023. The increase in average selling price reflected the impact of price increases enacted in 2023 and a change in the mix of products sold, as we strive to maximize capacity in our aerogel manufacturing facility. This increase in average selling price had the effect of increasing product revenue by \$9.9 million for the nine months ended September 30, 2024 from the comparable period in 2023.

In volume terms, energy industrial product shipments decreased by 15% as measured by square feet of our energy industrial products shipped for the nine months ended September 30, 2024, compared to the nine months ended September 30, 2023. The decrease in volume had the effect of decreasing product revenue by \$14.5 million for the nine months ended September 30, 2024 from the comparable period in 2023.

Thermal barrier revenue was \$236.8 million for the nine months ended September 30, 2024 as compared to \$57.2 million for the nine months ended September 30, 2023. During the nine months ended September 30, 2024 and 2023, thermal barrier revenue included \$223.8 million and \$49.0 million, respectively, to a major U.S. automotive OEM.

Cost of Revenue

	Nine Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Related Revenue	Amount	Percentage of Related Revenue	Amount	Percentage
Cost of revenue:						
Energy industrial	\$ 55,031	59%	\$ 73,065	75%	\$ (18,034)	(25)%
Thermal barrier	138,816	59%	54,131	95%	84,685	156%
Total cost of revenue	\$ 193,847	59%	\$ 127,196	82%	\$ 66,651	52%

Total cost of revenue increased \$66.6 million, or 52%, to \$193.8 million for the nine months ended September 30, 2024 from \$127.2 million in the comparable period in 2023. The increase in total cost of revenue was the result of an increase in thermal barrier cost of revenue, offset by a decrease in energy industrial cost of revenue.

Energy industrial cost of revenue decreased \$18.1 million, or 25%, to \$55.0 million for the nine months ended September 30, 2024 from \$73.1 million in the comparable period in 2023, primarily due to a decrease in the volume of products shipped and shift in manufacturing to the external manufacturing facility. The \$18.1 million decrease was the result of a \$2.0 million decrease in material costs and a \$16.1 million decrease in manufacturing and other operating costs from the comparable period in 2023.

Thermal barrier cost of revenue increased \$84.7 million to \$138.8 million for the nine months ended September 30, 2024 as compared to \$54.1 million in the comparable period in 2023. The \$84.7 million increase was the result of a \$63.2 million increase in material costs and a \$21.5 million increase in manufacturing costs, primarily driven by an increase in volume.

Gross Profit

	Nine Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
(\$ in thousands)						
Gross profit:						
Energy industrial	\$ 37,750	41%	\$ 24,246	25%	\$ 13,504	56%
Thermal barrier	98,014	41%	3,057	5%	94,957	(3106)%
Total gross profit (loss)	\$ 135,764	41%	\$ 27,303	18%	\$ 108,461	397%

Gross profit increased by \$108.5 million, or 397%, to \$135.8 million for the nine months ended September 30, 2024 from \$27.3 million of gross profit in the comparable period in 2023. The increase in gross profit was the result of the \$175.1 million increase in total revenue, offset by the \$66.6 million increase in total cost of revenue.

Research and Development Expenses

	Nine Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Research and development expenses	\$ 13,645	4%	\$ 12,281	8%	\$ 1,364	11%

Research and development expenses increased by \$1.3 million, or 11%, to \$13.6 million for the nine months ended September 30, 2024 from \$12.3 million in the comparable period in 2023. The \$1.3 million increase reflects increases in compensation and related costs of \$0.8 million, depreciation and facility related expenses of \$0.3 million, and other expenses of \$0.2 million.

Research and development expenses as a percentage of total revenue decreased to 4% of total revenue for the nine months ended September 30, 2024 from 8% in the comparable period in 2023.

Sales and Marketing Expenses

	Nine Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
Sales and marketing expenses	\$ 27,130	8%	\$ 24,226	16%	\$ 2,904	12%

Sales and marketing expenses increased by \$2.9 million, or 12%, to \$27.1 million for the nine months ended September 30, 2024 from \$24.2 million in the comparable period in 2023. The \$2.9 million increase was principally the result of increases in compensation and related costs of \$1.7 million, depreciation and facility related expenses of \$0.9 million, and utilities expenses of \$0.3 million.

Sales and marketing expenses as a percentage of total revenue decreased to 8% of total revenue for the nine months ended September 30, 2024 from 16% in the comparable period in 2023.

General and Administrative Expenses

	Nine Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
General and administrative expenses	\$ 52,465	16%	\$ 41,382	27%	\$ 11,083	27%

General and administrative expenses increased by \$11.1 million, or 27%, to \$52.5 million for the nine months ended September 30, 2024 from \$41.4 million in the comparable period in 2023. The \$11.1 million increase was the result of additional staffing costs combined with increases in compensation and related costs of \$4.5 million, foreign currency transaction losses of \$2.0 million, insurance costs of \$2.0 million, utilities expenditures of \$1.7 million, and legal costs of \$0.9 million. Compensation and related costs include \$2.0 million of charge from the cancellation of the unearned performance-based restricted shares.

General and administrative expenses as a percentage of total revenue decreased to 16% for the nine months ended September 30, 2024 from 27% in the comparable period in 2023.

Impairment of Equipment Under Development

The \$2.7 million impairment of equipment under development was the result of a charge for impairment of assets due to obsolescence following development of new and more efficient equipment.

Other Income (Expense), net

	Nine Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Interest income (expense):						
Interest (expense), related party	\$ (7,550)	(2)%	\$ (2,424)	(2)%	\$ (5,126)	211%
Interest income (expense), net	(883)	(0)%	5,532	4%	(6,415)	(116)%
Loss on extinguishment of debt	(27,487)	(8)%	—	—	(27,487)	NM
Income from Employee Retention Credits	—	—	2,186	1%	(2,186)	(100)%
Total interest income (expense), net	\$ (35,920)	(11)%	\$ 5,294	3%	\$ (41,214)	(779)%

Other income (expense), net decreased by \$41.2 million to \$35.9 million of other expense for the nine months ended September 30, 2024 from \$5.3 million of other income (expense), net in the comparable period in 2023. The \$41.2 million decrease was the result of a \$27.5 million loss on extinguishment of debt, \$6.1 million net impact of capitalized interest relating to our Convertible Note in the comparable period in 2023, \$1.8 million of deferred financing costs related to the GM Loan Agreement for which the draw down date has expired, a \$2.2 million income from Employee Retention Credits in the comparable period in 2023 not repeated in 2024, a \$1.9 million decrease of interest income, and a \$1.7 million increase of interest expense.

Income Tax Expense

	Nine Months Ended September 30,				Change	
	2024		2023			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
Income tax expense	\$ 1,889	1%	\$ —	0%	\$ 1,889	NM

The \$1.9 million of income tax expense for the nine months ended September 30, 2024 is related to our maquiladora operations in Mexico. We did not incur income tax expense for the comparable period in 2023.

Liquidity and Capital Resources

Overview

We have experienced significant losses and invested substantial resources since our inception to develop, commercialize and protect our aerogel technology and to build a manufacturing infrastructure capable of supplying aerogel products at the volumes and costs required by our customers. These investments have included research and development and other operating expenses, capital expenditures, and investment in working capital balances.

Our long-term financial projections anticipate revenue growth, increasing levels of gross profit, and improved cash flows from operations. To meet expected growth in demand for our aerogel products in the EV market, we have been in the process of expanding our aerogel blanket capacity by constructing a second manufacturing plant in Bulloch County, Georgia. However, in order to manage the development of the second plant so that its increased capacity comes online in a manner that aligns with our current expectations of demand from our EV customers, we are extending the timeframe for construction and commissioning of the second plant until such time as its capacity is supported by increased demand. In the meantime, and until we ramp up construction, we expect to be able to substantially reduce our planned capital expenditures for 2024. At the same time, we believe that productivity improvements in our existing Rhode Island facility combined with the supplemental supply of our energy industrial products from one or more external manufacturing facilities in China will permit us to achieve a target revenue capacity of approximately \$650 million in 2024 and prior to the completion and start-up of the second plant. Nonetheless, there can be no assurance as to when we will ramp up construction on the second plant. There can also be no assurance that our contract manufacturing strategy of meeting the demand of our energy industrial customers with supply from one or more external manufacturing facilities in China will provide us with adequate manufacturing capacity or supply for that expected demand. Furthermore, when we ramp up construction on the second plant, further cost inflation and/or supply chain disruptions, as well as potential changes in the scope of the facilities, could lead to increases to our prior estimated costs for completion of the second plant.

We are also increasing our investment in the research and development of next-generation aerogel products and technologies. During 2024, we have and will continue to develop aerogel products and technologies for the EV market. We believe the commercial potential for our technology in the EV market is significant. To meet the anticipated revenue growth and take advantage of this market opportunity, we are adding personnel and incurring additional operating expenses, among other items.

In January and September 2024, we entered into sale and leaseback arrangements, pursuant to which we sold certain equipment to an equipment leasing company for two separate one-time cash payments of \$5.0 million and \$10.0 million and leased back such equipment from the leasing company. The associated monthly lease rents will be paid over the term of three years.

In October 2024, we entered into an underwriting agreement (the Underwriting Agreement) with Goldman Sachs & Co. LLC and Morgan Stanley & Co. LLC (collectively, the Underwriters), pursuant to which we issued and sold an aggregate of 4,887,500 shares of our common stock, which included 637,500 shares pursuant to the Underwriters' option to purchase additional shares of our common stock, to the Underwriters in a registered underwritten offering (the "Offering"). The price to the public in the Offering was \$20.00 per share. The net proceeds to us from the Offering were approximately \$93.2 million, after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

On August 19, 2024, we entered into a Credit, Security and Guaranty Agreement with MidCap Financial for a term loan facility in an aggregate principal amount of \$125.0 million and an asset-based revolving credit facility in an aggregate principal amount not to exceed the lesser of \$100.0 million or the borrowing base. At closing we drew \$125.0 million from the Term Loan Facility and \$43.0 million from the revolving loan facility. The proceeds of the borrowings, net of fees and costs, were used for repurchasing of the Convertible Notes for \$150.0 million and for general corporate purposes.

We believe that our September 30, 2024 cash and cash equivalents balance of \$113.5 million, together with the funds available under the MidCap Loan Facility, will be sufficient to support current operating requirements, current research and development activities and the initial capital expenditures required to support the evolving commercial opportunities in the EV market and other strategic business opportunities.

We plan to supplement our cash balance and available credit with equity financings, debt financings, equipment leasing, customer prepayments or government grant and loan programs to provide the additional capital necessary to purchase the capital equipment, construct the new facilities and complete the aerogel capacity expansions required to support our evolving commercial opportunities and strategic business initiatives.

We believe that consummation of equity financing could potentially result in an ownership change under Section 382 of the U.S. Internal Revenue Code of 1986, as amended. Such an ownership change would lead to the use of our net operating loss carryforwards being restricted. Our inability to use a substantial portion of our net operating loss carryforwards would result in a higher effective tax rate and adversely affect our financial condition and results of operations.

Primary Sources of Liquidity

Our principal sources of liquidity are currently our cash and cash equivalents. Cash and cash equivalents consist primarily of cash, money market accounts, and sweep accounts on deposit with banks. As of September 30, 2024, we had \$113.5 million of unrestricted cash and cash equivalents.

Analysis of Cash Flow

Net Cash Used in Operating Activities

During the nine months ended September 30, 2024, we provided \$9.9 million in net cash in operating activities, as compared to the use of \$39.8 million in net cash during the comparable period in 2023, a decrease in the use of cash of \$49.7 million. This decrease in use of cash was the result of net income adjusted for non-cash items of \$97.8 million offset by net cash used by changes in operating assets and liabilities of \$48.1 million.

During the nine months ended September 30, 2023, we used \$39.8 million in net cash in operating activities, as compared to the use of \$70.3 million in net cash during the comparable period in 2022, a decrease in the use of cash of \$30.5 million. This decrease in use of cash was the result of lower net loss adjusted for non-cash items of \$31.4 million offset by net cash used by changes in operating assets and liabilities of \$0.9 million.

Net Cash Used in Investing Activities

Net cash used in investing activities is for capital expenditures for machinery and equipment principally to improve the throughput, efficiency and capacity of our East Providence facility, expansion of our automated fabrication facility in Mexico and engineering designs and construction costs for the planned aerogel manufacturing facility in Bulloch County, Georgia. Net cash used in investing activities for the nine months ended September 30, 2024 and 2023 was \$71.5 million and \$147.7 million, respectively.

Net Cash Provided by Financing Activities

Net cash provided by financing activities for the nine months ended September 30, 2024 totaled \$35.6 million and consisted of \$15.0 million in proceeds from a sales leaseback, \$10.4 million in proceeds from employee stock option exercises, \$119.7 in proceeds from the term loan, net of issues costs, and \$42.7 in proceeds from the revolver, net of issuance costs, offset by \$150.0 million in cash used for the repayment of the convertible note, \$1.3 million in cash used for payments made for employee tax withholdings associated with the vesting of restricted stock units, \$0.8 million in repayments of a sales leaseback, and less than \$0.1 million in cash used for fees and issuance costs.

Net cash used in financing activities for the nine months ended September 30, 2023 totaled \$0.1 million and consisted of \$0.4 million in cash used for payments made for employee tax withholdings associated with the vesting of restricted stock units and \$0.1 million issuance costs from private placement of common stock, offset, in part, by \$0.4 million in proceeds from employee stock option exercises.

Contractual Obligations and Commitments

There have been no material changes to our contractual obligations and commitments as reported in our Annual Report.

Recent Accounting Pronouncements

Information regarding new accounting pronouncements is included in note 2 to our unaudited consolidated financial statements contained in Item 1 of this Quarterly Report on Form 10-Q.

Critical Accounting Policies and Estimates

Our financial statements are prepared in accordance with U.S. GAAP. The preparation of our financial statements and related disclosures requires us to make estimates, assumptions and judgments that affect the reported amount of assets, liabilities, revenue, costs and expenses and related disclosures. We believe that the estimates, assumptions and judgments involved in these accounting policies have the greatest potential impact on our financial statements and, therefore, we consider these to be our critical accounting policies. Accordingly, we evaluate our estimates and assumptions on an ongoing basis. Our actual results may differ from these estimates under different assumptions and conditions. See our Annual Report and note 2 to our consolidated financial statements included elsewhere in this Quarterly Report on Form 10-Q for information about these critical accounting policies, as well as a description of our other significant accounting policies.

Certain Factors That May Affect Future Results of Operations

The SEC encourages companies to disclose forward-looking information so that investors can better understand a company's future prospects and make informed investment decisions. This Quarterly Report on Form 10-Q contains such "forward-looking statements" within the meaning of the Private Securities Litigation Reform Act of 1995. These statements involve known and unknown risks, uncertainties and other important factors, which may cause our actual results, performance or achievements to be materially different from any future results, performances or achievements expressed or implied by the forward-looking statements. Forward-looking statements include, but are not limited to, statements about: the expected future growth of the market for our aerogel products and our continued gain in market share, in particular in the electric vehicle market, the energy infrastructure insulation market, the lithium-ion battery thermal barrier markets, and other markets we target; our beliefs in the appropriateness of our assumptions, the accuracy of our estimates regarding expenses, loss contingencies, future revenues, revenue capacity, future profits, uses of cash, available credit, capital requirements, and the need for additional financing to operate our business and for capital expenditures and to fund our planned strategic business initiatives; the performance of our aerogel blankets; our expectation that we will be successful in obtaining, enforcing and defending our patents against competitors and that such patents are valid and enforceable; our expectations regarding the investment and revenue creation from the Statesboro Plant and creation of jobs from the construction and operation of the Statesboro Plant; ; our efforts to manage the extended construction and commission timeframe of the planned Statesboro Plant to align with our expectations of demand from EV customers; our estimates of annual production capacity; beliefs about the commercial potential for our technology in the electric vehicle market; beliefs about our ability to produce and deliver products to electric vehicle customers; beliefs about our contracts with the major automotive manufacturers; our expectations about the size and timing of awarded business in the electric vehicle market, future revenues and profit margins, arising from our supply relationship and contract with automotive OEMs and our ability to win more business and increase revenue in the electric vehicle market; beliefs about the performance of our thermal barrier products in the battery systems of electric vehicles; the current or future trends in the energy, energy infrastructure, chemical and refinery, LNG, sustainable building materials, electric vehicle thermal barrier, electric vehicle battery materials or other markets and the impact of these trends on our business; our investments in the electric vehicle market and Aerogel Technology Platform; our beliefs about the financial metrics that are indicative of our core performance; our expectations about the effect of manufacturing capacity on financial metrics such as Adjusted EBITDA; our expectations about future revenues, expenses, gross profit, net income (loss), income (loss) per share and Adjusted EBITDA, sources and uses of cash, capital requirements and the sufficiency of our existing cash balance and available credit; our beliefs about the outcome, effects or estimated costs of current or potential litigation or their respective timing, including expected legal expense in connection with our patent enforcement actions; our expectations about future material costs and manufacturing expenses as a percentage of revenue, including the impact of engaging one or more external manufacturing facilities in China for supply of our energy industrial products; our expectation about the ability of the Chinese external manufacturing facilities that we engage to consistently supply the aerogel product that we order in a timely manner; our expectations of future gross profit and the effect of manufacturing expenses, manufacturing capacity and productivity on gross profit; our expectations about our resources and other investments in new technology and related research and development activities and associated expenses; our expectations about short and long term (a) research and development (b) general and administrative and (c) sales and marketing expenses; our expectations of revenue growth, increased gross profit, and improving cash flows over the long term; our intentions about managing capital

expenditures and working capital balances; our expectation with respect to the receipt and funding of the Loan, including satisfaction of certain technical, legal, environmental, and financial conditions before the DOE enters into definitive financing documents and funds the Loan; our expectation with respect to the ability of the DOE loan to fully fund the remaining CAPEX needed to complete the Statesboro Plant; our expectations and beliefs regarding the use of proceeds from the Loan facility, the flexibility and efficiency of the facility; and our expectations about potential sources of future financing.

Words such as “may,” “will,” “anticipate,” “estimate,” “expects,” “projects,” “intends,” “plans,” “believes” and words and terms of similar substance used in connection with any discussion of future operating or financial performance, identify forward-looking statements. All forward-looking statements are management’s present expectations of future events and are subject to a number of risks and uncertainties that could cause actual results to differ materially and adversely from those described in the forward-looking statements. These risks include, but are not limited to, those set forth in this Quarterly Report on Form 10-Q and under the heading “Risk Factors” contained in Item 1A of our Annual Report.

In light of these assumptions, risks and uncertainties, the results and events discussed in the forward-looking statements contained in this Quarterly Report on Form 10-Q might not occur. Stockholders and other readers are cautioned not to place undue reliance on the forward-looking statements, which speak only as of the date of this Quarterly Report on Form 10-Q. We are not under any obligation, and we expressly disclaim any obligation, to update or alter any forward-looking statements, whether as a result of new information, future events or otherwise. All subsequent forward-looking statements attributable to Aspen Aerogels, Inc. or to any person acting on its behalf are expressly qualified in their entirety by the cautionary statements contained or referred to in this section.

Item 3. Quantitative and Qualitative Disclosures About Market Risk.

Market risk represents the risk of loss that may impact our financial position due to adverse changes in financial market prices and rates. Our market risk exposure results primarily from fluctuations in interest rates, as well as from inflation. In the normal course of business, we are exposed to market risks, including changes in interest rates which affect our cash flows. We may also face additional exchange rate risk in the future as we expand our business internationally.

Interest Rate Risk

We are exposed to changes in interest rates in the normal course of our business. As of September 30, 2024, we had unrestricted cash and cash equivalents of \$113.5 million. These amounts were held for working capital and capital expansion purposes and were invested primarily in deposit accounts, money market accounts, and high-quality debt securities issued by the U.S. government via cash sweep accounts primarily at major financial institutions in North America. Due to the short-term nature of these investments, we believe that our exposure to changes in the fair value of our cash as a result of changes in interest rates is not material.

As of September 30, 2024, we had a term loan outstanding with principal balance of \$127.5 million. Our term loan bears interest at the Secured Overnight Financing Rate (SOFR) plus 4.50% per annum. Under the terms of the investment, SOFR has a floor of 4.50% and a cap of 7.50%. Interest is paid quarterly.

As of September 30, 2024, we had a revolving facility outstanding with principal balance of \$43.0 million. Our revolving facility bears interest at the SOFR plus 4.60% per annum. Under the terms of the investment, SOFR has a floor of 2.50%. Interest is paid monthly. The amount available to us under the revolving credit facility was \$25.7 million.

As of September 30, 2024, we had \$0.4 million of restricted cash to support our outstanding letters of credit to secure obligations under certain commercial contracts and other obligations. We terminated our revolving credit facility agreement on November 28, 2022.

Inflation Risk

Although we expect that our operating results will be influenced by general economic conditions, we do not believe that inflation has had a material effect on our results of operations during the periods presented in this report. However, our business may be affected by inflation in the future.

Foreign Currency Exchange Risk

We are subject to inherent risks attributed to operating in a global economy. We do not consider the exposure to foreign currency risk from our international operations to be material. A majority of our revenue, receivables, purchases and debts are denominated in U.S. dollars. Certain transactions of the Company and its subsidiaries are denominated in currencies other than the functional currency. Foreign currency transaction losses were \$2.3 million and \$0.3 million for the nine months ended September 30, 2024, and 2023, respectively, and were recorded within operating expenses on the consolidated statements of operations.

Item 4. Controls and Procedures.***(a) Evaluation of Disclosure Controls and Procedures.***

We maintain disclosure controls and procedures that are designed to ensure that information required to be disclosed in the reports that we file or submit under the Securities Exchange Act of 1934, as amended (the Exchange Act), is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms and is accumulated and communicated to our management, including our principal executive officer and principal financial officer, or persons performing similar functions, as appropriate to allow timely decisions regarding required disclosure.

As of September 30, 2024, our management, with the participation of our principal executive officer and principal financial officer, evaluated the effectiveness of our disclosure controls and procedures (as defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act). Our management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving their objectives, and management necessarily applies its judgment in evaluating the cost-benefit relationship of possible controls and procedures. Based on such evaluation, our principal executive officer and principal

financial officer have concluded that, as of September 30, 2024, our disclosure controls and procedures were effective to ensure that information required to be disclosed by us in the reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms, and is accumulated and communicated to our management, including our principal executive officer and principal financial officer, or persons performing similar functions, as appropriate to allow timely decisions regarding required disclosure.

(b) Changes in Internal Controls.

During the three months ended September 30, 2024, there were no changes in our internal control over financial reporting, as such term is defined in Rules 13a-15(f) and 15(d)-15(f) promulgated under the Exchange Act, that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

PART II — OTHER INFORMATION

Item 1. Legal Proceedings.

We are involved in various legal claims and proceedings in the normal course of operations. We believe the outcome of these matters will not have a material adverse effect on our consolidated financial position, results of operations or liquidity, except as described in Part 1, Item 3. “Legal Proceedings” of our Annual Report on Form 10-K. Since the filing of our Form 10-K, there have been no material changes in our legal proceedings from those disclosed therein, other than as noted below.

Our patent infringement proceedings in Italy against of AMA S.p.A. and AMA Composites S.r.l. (collectively, AMA) are ongoing. In July 2023, the technical experts appointed by the judge issued a report finding key claims of our process patents valid and infringed by the aerogel products manufactured by Nano Tech Co., Ltd. and sold by AMA. In November 2023, we appealed our motion for preliminary injunction to a panel of the Court of Genoa. In February 2024, the Court issued an order dismissing our appeal. As of March 31, 2024, the main patent infringement proceedings are ongoing.

Our patent infringement proceedings in Korea against Beerenberg Services AS, Beerenberg Korea Ltd., and Bronx (China) Co., Ltd., are ongoing. In April 2024, the Korea Trade Commission (KTC) concluded its investigation into unfair international trade practices by Beerenberg Korea Ltd. and Bronx (China) Co., Ltd., with a decision that our asserted composition patent claims were invalid and our asserted process patent claims were not infringed. We disagree with the KTC decision, which is subject to appeal in the Seoul Administrative Court. We continue to vigorously defend the validity of these patents in the related oppositions filed by Beerenberg Korea Ltd. and Bronx (China) Co. Ltd., at the Korean Intellectual Property Trial and Appeal Board (IPTAB). Our patent infringement case against Beerenberg Services AS and Beerenberg Korea Ltd. at the Seoul District Court remains stayed pending the outcome of the IPTAB proceedings.

The opposition filed in September 2023 by LG Chem Ltd. against one of the Korean patents we are asserting against Beerenberg in Korea is ongoing. We are vigorously defending the validity of this patent. The opposition filed in August 2023 by LG Chem Ltd. against a Japanese counterpart of the Korean patents was concluded in our favor in May 2023 with a decision to maintain the patent and issuance of new Certificate of Patent on May 3, 2024.

In October 2022, we were served with a summons from Aerogels Poland Nanotechnology LLC (APN), a former distributor of our products in Poland with whom we previously terminated our distribution agreements because of APN’s failure to pay amounts due to us. The summons asserts causes of action for declaratory judgment, breach of contract, breach of implied contract, equitable estoppel and fraud, and states that plaintiffs will seek declaratory judgment, actual and liquidated damages in the sum of \$20 million, in addition to attorneys’ fees. We were not served with any complaint at the time the summons was served. In December 2022, we filed a notice of appearance in New York County Supreme Court and a demand upon plaintiffs to file and serve a complaint. In March 2023, plaintiffs filed a complaint asserting various causes of action consistent with those set forth in the October 2022 summons, and a demand for monetary damages and other relief in excess of \$16 million. In July 2023, we filed a motion to compel arbitration, and in February 2024, the Court granted our motion and stayed the litigation pending arbitration. To the extent APN seeks to pursue claims in an arbitration proceeding, Aspen intends to continue to vigorously defend this matter, including seeking its legal costs.

Item 1A. Risk Factors.

The ownership of our common stock involves a number of risks and uncertainties. When evaluating the Company and our business before making an investment decision regarding our securities, potential investors should carefully consider the risk factors and uncertainties described in Part 1, Item 1A. “Risk Factors” of our Annual Report on Form 10-K. Since the filing of our Form 10-K, there have been no material changes in our risk factors from those disclosed therein, other than as provided below.

We are dependent upon our ability to close on and fully draw down on our proposed loan facility from the United States Department of Energy, which may restrict our ability to conduct our business.

The receipt by us of the conditional commitment from the DOE in respect of the Loan is not an assurance that definitive financing documents will be negotiated and entered into, that the terms and conditions of the Loan will be consistent with the terms contemplated in the conditional commitment, or that all technical, legal, environmental and financial conditions will be achieved and satisfied. The outcome of the process with the DOE relating to the Loan and our ability to draw and utilize such funds towards the construction of the Statesboro Plant is dependent on whether the DOE determines to proceed with the definitive financing documents and under what conditions. There can be no assurance as to the outcome of these decisions and the terms and conditions of the financing documents that may be offered, if any.

If and when definitive financing documents in respect of the Loan are entered into, the Loan and our ability to draw upon the Loan are expected to be conditional upon certain condition precedents and to include representations, warranties and covenants of us customary for financings of a similar nature by the U.S. Government or other public lending institutions, including relating to repayment obligations. Our failure to comply with or satisfy any or all of the conditions or to remain in compliance with the covenant regime under the Loan could result in a reduction of the size of the Loan made available to us and/or an event of default under the Loan, which could result in the termination of the Loan and/or cause any amounts outstanding under the Loan to become immediately due and payable, any of which would have a material adverse effect on our business, financial condition, results of operations and prospects.

We have engaged third-party external manufacturing facilities in China to supplement our supply of our energy industrial products. If such external manufacturing facilities are unable to manufacture and deliver a sufficient quantity of high-quality products on a timely and cost-efficient basis, our net revenue and business operations may be harmed and our reputation may suffer.

We have engaged external manufacturing facilities in China for a supplemental supply of our energy industrial products, which we believe will enable us to achieve a target revenue capacity of approximately \$650 million in the near-term and prior to the completion and start-up of our planned second manufacturing plant. If our external manufacturing facilities are unable to deliver the required aerogel product on a timely basis, we may experience delays in delivering our finished aerogel product to customers in the energy industrial market. In addition, because our third-party external manufacturing facilities have manufacturing facilities in China, their ability to provide us with adequate supplies of high-quality products on a timely and cost-efficient basis is subject to a number of additional risks and uncertainties, including political, social and economic instability and other factors that could impact the shipment of supplies. In 2022, the indirect parent of the external manufacturing facility we engaged in 2023 was added to the list of “Chinese military companies” that are “operating directly or indirectly in the United States” in accordance with Section 1260H of the National Defense Authorization Act for Fiscal Year 2021. There are certain government contract related restrictions that are or will be imposed on Section 1260H list entities and their controlled affiliates. While we currently have no contracts that would be affected by these restrictions, and the legal impact of being included on the list is relatively limited, our work with a listed entity may have a material adverse effect on our reputation and our business opportunities. If our manufacturers are unable to provide us with adequate supplies of high-quality aerogel products on a timely and cost-efficient basis, our operations could be disrupted and our revenue and business operations may suffer. Moreover, if our third-party external manufacturing facilities cannot consistently produce high-quality products that are free of defects and/or violates applicable worker or product safety rules, regulations or laws, we may experience a loss of customers, which may also reduce our revenues and may harm our reputation and brand. Furthermore, our third-party external manufacturing facilities may become subject to various supply chain disruptions, including for key inputs into their manufacturing process such as methanol. Such disruptions could be the result of pandemics or public health crises, and/or geopolitical disputes and conflicts, any of which could slow or halt the delivery of products to us and increase the price of certain materials due to resulting increases in costs of raw materials and shipping costs.

At present, we have engaged one third-party external manufacturing facility in China to produce certain of our Energy Industrial products. Pursuant to our supply contract with this contract manufacturer, they are obligated to deliver products to us as we issue purchase orders on an as-needed basis through the term of the agreement. The contract automatically renews year-to-year unless either party notifies the other of its intention not to renew the contract. While we have agreed to purchase our requirement for certain Energy Industrial products from the contract manufacturer, we have no obligation to purchase any minimum quantity under the contract. In addition, we may terminate the contract at any time and for any or no reason.

Item 2. Unregistered Sales of Equity Securities and Use of Proceeds.

(a) Unregistered Sales of Equity Securities.

None.

(b) Use of Proceeds from Initial Public Offering of Common Stock.

Not applicable.

(c) Purchases of Equity Securities by the Issuer and Affiliated Purchasers.

We did not repurchase any of our equity securities during the quarter ended September 30, 2024.

Item 3. Defaults Upon Senior Securities.

None.

Item 4. Mine Safety Disclosures.

Not applicable.

Item 5. Other Information.

None of our directors or executive officers adopted, modified or terminated any contract, instruction or written plan for the purchase or sale of our securities that was intended to satisfy the affirmative defense conditions of Rule 10b5-1(c) or any “non-Rule 10b5-1 trading arrangement” as such term is defined in Item 408(a) of Regulation S-K, during the fiscal quarter ended September 30, 2024.

Item 6. Exhibits.

(a) Exhibits

10.1*	<u>Note Purchase and Sale Agreement, dated as of August 19, 2024, by and between Aspen Aerogels, Inc. and Wood River Capital, LLC (incorporated by reference to Exhibit 10.1 of the Company's Current Report on Form 8-K, filed with SEC on August 19, 2024).</u>
10.2*	<u>Credit, Security and Guaranty Agreement, dated as of August 19, 2024, by and among Aspen Aerogels, Inc., Aspen Aerogels Rhode Island, LLC, MidCap Fund IV Trust and MidCap Financial Trust (incorporated by reference to Exhibit 10.2 of the Company's Current Report on Form 8-K, filed with SEC on August 19, 2024).</u>
10.3+	<u>Amended and Restated Executive Employment Agreement, dated as of August 30, 2024, by and between the Company and Donald R. Young (incorporated by reference to Exhibit 10.1 of the Company's Current Report on Form 8-K, filed with SEC on September 3, 2024).</u>
31.1	<u>Certification of principal executive officer under Section 302(a) of the Sarbanes-Oxley Act of 2002.</u>
31.2	<u>Certification of principal financial officer under Section 302(a) of the Sarbanes-Oxley Act of 2002.</u>
32	<u>Certifications of the principal executive officer and the principal financial officer under Section 906 of the Sarbanes-Oxley Act of 2002.</u>
101.INS	XBRL Instance Document - the instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document.
101.SCH	Inline XBRL Taxonomy Extension Schema With Embedded Linkbase Documents.
104	Cover Page Interactive Data File (embedded within the Inline XBRL document).

* Certain exhibits and schedules have been omitted in accordance with Item 601(a)(5) of Regulation S-K. The registrant agrees to furnish supplementally a copy of any omitted exhibit or schedule to the SEC upon its request.

+ Management contract or compensatory plan or arrangement.

SIGNATURES

Pursuant to the requirements of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned thereunto duly authorized.

ASPEN AEROGELS, INC.

Date: November 7, 2024

By: /s/ Donald R. Young
Donald R. Young
President and Chief Executive Officer
(principal executive officer)

Date: November 7, 2024

By: /s/ Ricardo C. Rodriguez
Ricardo C. Rodriguez
Chief Financial Officer and Treasurer
(principal financial officer)

CERTIFICATIONS UNDER SECTION 302

I, Donald R. Young, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Aspen Aerogels, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: November 7, 2024

/s/ Donald R. Young

Donald R. Young
President and Chief Executive Officer
(principal executive officer)

CERTIFICATIONS UNDER SECTION 302

I, Ricardo C. Rodriguez, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Aspen Aerogels, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: November 7, 2024

/s/ Ricardo C. Rodriguez

Ricardo C. Rodriguez

Chief Financial Officer and Treasurer (principal financial officer)

CERTIFICATIONS UNDER SECTION 906

Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002 (subsections (a) and (b) of section 1350, chapter 63 of title 18, United States Code), each of the undersigned officers of Aspen Aerogels, Inc., a Delaware corporation (the “Company”), does hereby certify, to such officer’s knowledge, that:

The Quarterly Report on Form 10-Q for the quarter ended September 30, 2024 (the “Form 10-Q”) of the Company fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934, and the information contained in the Form 10-Q fairly presents, in all material respects, the financial condition and results of operations of the Company.

Dated: November 7, 2024

/s/ Donald R. Young

Donald R. Young
President and Chief Executive Officer
(principal executive officer)

Dated: November 7, 2024

/s/ Ricardo C. Rodriguez

Ricardo C. Rodriguez
Chief Financial Officer and Treasurer
(principal financial officer)

A signed original of this written statement required by Section 906 has been provided to the Company and will be retained by the Company and furnished to the Securities and Exchange Commission or its staff upon request.
